Ministry of High Education

Culture and Science city at Oct 6,

The High Institute of Computer Science & Information Systems

Graduation Project

Smart Medication Management System for Elderly People

A graduation project submitted in partial fulfilment of the requirements for the degree of Bachelor of Computer Science and Information System.

Supervised by:

Dr. Eng. Omaima Goher

Assistant

TA. Menna Magdy

Project No: F26_504 **Academic Year: 2025/ 2026**

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Smart Medication Management System for Elderly People

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Chapter 1: Introduction

1.1 Overview

The rapid growth of the elderly population has increased the demand for effective and accessible healthcare management solutions. Elderly patients often encounter significant difficulties in adhering to prescribed medication schedules and in monitoring essential health indicators such as blood pressure and blood glucose levels. Conventional healthcare follow-up methods frequently rely on repeated clinical visits, which may impose physical, logistical, and psychological burdens on patients. In this context, recent advances in mobile technology and cloud computing have created valuable opportunities for improving healthcare delivery and strengthening communication between patients and healthcare providers. The proposed Smart Medication Management System for Elderly People is designed as a mobile health solution that supports secure data storage, continuous accessibility, and efficient data sharing, thereby contributing to improved patient engagement and more effective healthcare management.

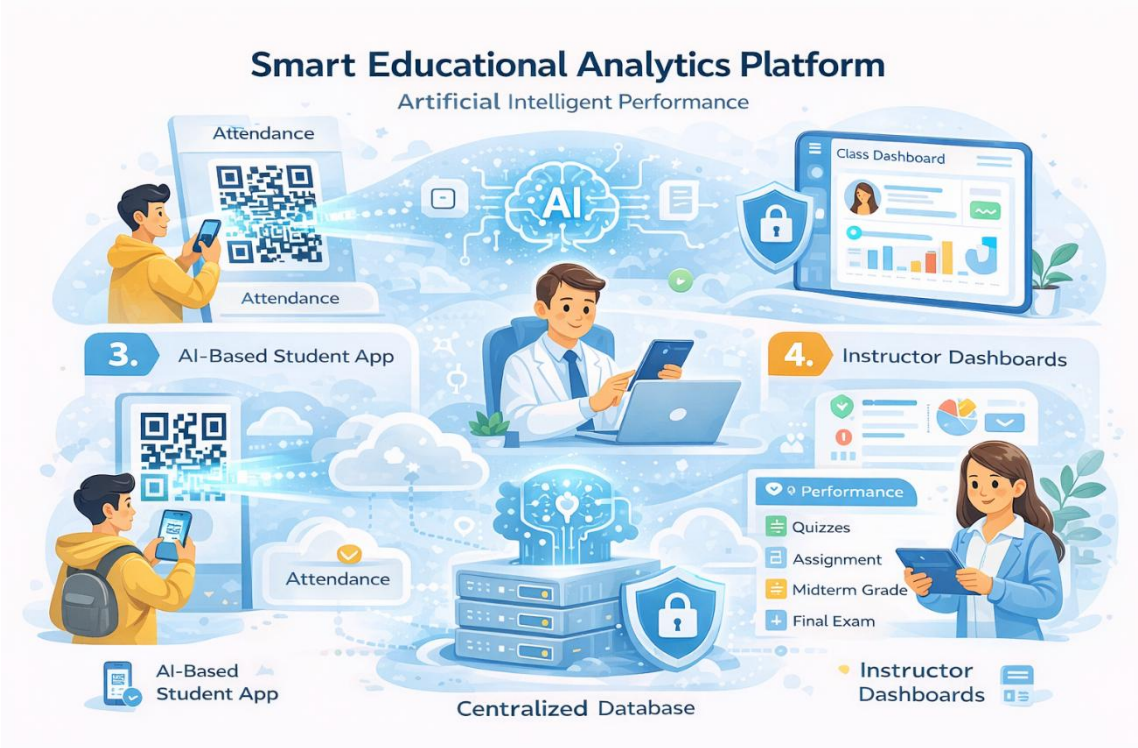


Figure 1 General Overview of the Process

1.2 Motivation

The motivation for this project stems from the persistent challenges associated with medication non-adherence and incomplete health monitoring among elderly patients. Many patients experience difficulty in maintaining consistent medication routines because of memory-related limitations and the complexity of treatment schedules. At the same time, healthcare providers may not always have immediate access to up-to-date patient information, which reduces the ability to monitor treatment progress and respond promptly when intervention is needed. Traditional healthcare systems, which depend heavily on face-to-face visits, may not sufficiently address these issues in a timely and practical manner. Therefore, there is a clear need for an integrated digital system capable of supporting medication adherence, facilitating health data recording, and improving real-time interaction between patients and healthcare professionals.

1.3 Objective

The primary objective of this project is to design and develop a mobile application that assists patients in following their medication schedules accurately through an effective reminder mechanism. In addition, the system aims to enable users to record important vital health indicators, including blood pressure and glucose levels, while providing healthcare professionals with secure cloud-based access to patient data in real time. The project also seeks to facilitate quick and practical data sharing through QR code functionality, improve usability through an intuitive interface suitable for elderly users, and ensure the privacy and security of sensitive medical information. Furthermore, the system is intended to provide a scalable foundation that can support future extensions such as wearable-device integration, telemedicine services, and drug interaction verification.

1.4 Aim

The aim of this project is to enhance the quality of life of elderly patients by providing a reliable, secure, and user-friendly digital platform for medication management and health monitoring. The system is intended to reduce medication errors, encourage timely adherence to prescribed treatment plans, and support the regular recording of essential health measurements. In parallel, it aims to assist healthcare providers by offering timely access to patient information, thereby improving follow-up efficiency, supporting early intervention, and promoting safer and more informed clinical decision-making.

1.5 Scope

The scope of the Smart Medication Management System focuses primarily on elderly patients, while remaining applicable to users from other age groups. In its current phase, the system is designed to provide three principal functions: medication reminders, health data recording, and QR code-based data sharing. The application relies on cloud technology to ensure secure storage and efficient management of patient information, while also supporting multilingual interfaces to address diverse user needs. In addition, the system includes a backend structure capable of handling health-related data efficiently. The project scope also recognizes future enhancement opportunities, including remote monitoring capabilities and drug interaction checking, which may further extend the system's usefulness and effectiveness in subsequent development phases.



Figure 2 Scope of the Smart Medication Management System

1.6 General Constraints

The development and implementation of the Smart Medication Management System are influenced by several general constraints. First, the current version of the application is centered on core functionalities such as authentication, medication management, health readings tracking, bilingual support, cloud synchronization, and essential security features, while several advanced functions remain planned for future implementation. Second, the present deployment emphasis is on Android devices, with specified minimum system and hardware requirements, whereas iOS support is considered as part of the application's future expansion. Third, although the system supports offline data persistence, full real-time synchronization depends on the availability of internet connectivity. Finally, some optional features—such as biometric authentication, camera usage, and other device-dependent capabilities—are limited by the technical specifications of the user's.

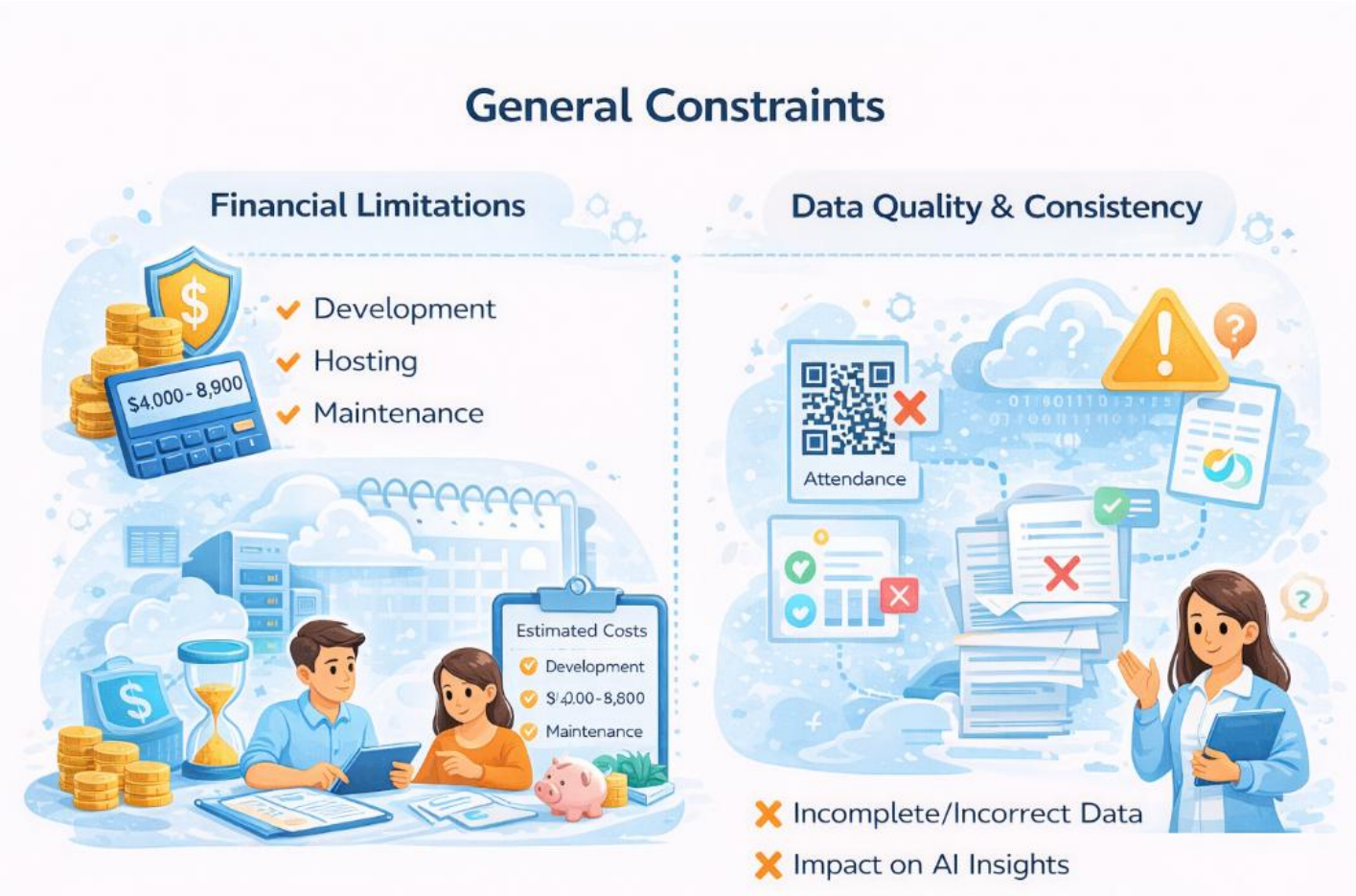


Figure 3 General Constraints of the Smart Medication Management System

1.7 Organization of the Project

The application is structured using a modular, Feature-Driven Architecture to ensure scalability, high cohesion, and a clear separation of concerns. The codebase, located primarily in the lib/ directory, is organized into the following main components:

Core (lib/core/): Serves as the foundational layer. It contains shared application resources, including the design system (themes, typography, colors), global reusable widgets, utility functions, error handling mechanisms, and core application-level configurations.

Features (lib/features/): The heart of the application, organized by business logic rather than technical layers. Each feature (e.g., auth, medicines, doctor_dashboard, health_readings) encapsulates its own screens, state management (Providers), and specific logic, making the code highly maintainable and isolated.

Services (lib/services/): Acts as the communication layer for external integrations, handling operations like Firebase Authentication, Cloud Messaging (FCM), and database interactions.

Entry Point (lib/main.dart): The starting point of the application. It handles a strict initialization sequence—configuring Firebase, local databases, device security checks, and notification services—before launching the user interface.

Chapter 2 Background and Previous Work

2.1 Background

The healthcare sector has witnessed a significant digital transformation in recent years, especially with the increasing use of mobile health applications to support patient care, chronic disease management, and medication adherence. Mobile health applications allow patients to manage their medical information, receive medication reminders, record health readings, and communicate more effectively with healthcare providers.

Medication adherence is one of the most important challenges in healthcare, particularly among elderly patients and patients with chronic diseases. Elderly patients often take multiple medications daily, which increases the risk of forgetting doses, taking medication at the wrong time, or failing to follow the prescribed treatment plan correctly. Poor medication adherence can lead to treatment failure, disease progression, increased hospital admissions, higher healthcare costs, and reduced quality of life. The World Health Organization reported that adherence to long-term therapies in developed countries averages only about 50%, and that the percentage can be lower in developing countries [1].

In addition to medication adherence, elderly patients often need to monitor vital health indicators such as blood pressure, blood glucose level, weight, and temperature. Recording these measurements over time helps patients and doctors track health changes, detect abnormal patterns, and make better clinical decisions.

The proposed **Smart Medication Management System for Elderly People** is designed to address these challenges by providing a mobile-based solution that combines medication reminders, health readings tracking, cloud-based data storage, QR-code-based data sharing, bilingual support, and secure access for both patients and doctors. The system aims to improve medication adherence, support continuous health monitoring, and enhance communication between elderly patients and healthcare professionals.

2.2 Previous Work

Previous research has shown that mobile health applications can play an important role in improving medication adherence and supporting patient self-management. Medication reminder applications provide patients with scheduled alerts, medication logs, refill reminders, and sometimes health tracking tools. These features help reduce dependence on memory and support patients in following their treatment plans more consistently.

Santo et al. conducted a systematic review of medication reminder applications and identified a large number of available apps designed to improve medication adherence. The study found that common features included flexible scheduling, medication tracking history, snooze options, and visual aids. However, the review also showed that many applications included only a limited number of desirable features, which indicates the need for more complete and integrated solutions.

Recent studies also indicate that mobile applications can improve medication adherence among patients with chronic conditions. A systematic review and meta-analysis published in the Journal of Medical Internet Research reported that mobile app interventions improved medication adherence across the included randomized controlled trials, with most studies showing statistically significant improvement.

Existing commercial applications also provide useful medication management features. MyTherapy provides medication reminders, intake documentation, refill reminders, health measurements tracking, symptom tracking, and health reports for doctor visits. Medisafe supports medication reminders, complex dosing schedules, caregiver support, drug interaction warnings, refill reminders, and measurement tracking. CareClinic provides broader health tracking features, including medication tracking, symptoms, health records, blood pressure, blood sugar, mood tracking, and reports. Dosecast focuses strongly on medication reminders, flexible scheduling, dose history, refill alerts, and reminders that can work without an internet connection.

Although these systems provide valuable features, many of them do not fully combine elderly-oriented design, Arabic-English localization, medication reminders, health readings, QR-code-based doctor access, offline support, and role-based patient-doctor security in one integrated solution. Therefore, the Smart Medication Management System builds on the strengths of previous work while addressing these limitations through a unified mobile healthcare platform.

2.3 Literature Review

2.3.1 Medication Non-Adherence among Elderly Patients

The World Health Organization reported that adherence to long-term therapies in developed countries averages only about 50%, and that the percentage can be lower in developing countries [1].

Elderly patients are more vulnerable to medication non-adherence because they often suffer from multiple chronic conditions and may need to take several medications daily. This increases the complexity of treatment schedules and makes it harder for patients to remember the correct medicine, dosage, and time. Factors such as forgetfulness, cognitive decline, poor health literacy, side effects, and lack of communication with healthcare providers can further reduce adherence.

Medication non-adherence can lead to serious consequences, including poor disease control, increased hospital admissions, higher healthcare costs, reduced quality of life, and increased morbidity and mortality. Therefore, improving adherence is considered an important goal in chronic disease management and elderly healthcare.

2.3.2 Mobile Health Applications for Medication Management

Mobile health, or mHealth, refers to the use of mobile devices and wireless technologies to support healthcare services. In recent years, mHealth applications have become increasingly important in medication management because they provide patients with reminders, tracking tools, educational content, and communication support. These applications can help patients become more involved in their treatment and reduce dependence on memory-based medication routines.

Several studies have shown that mobile applications can improve medication adherence among patients with chronic diseases. A systematic review published in the Journal of Medical Internet Research reported that mobile applications can help increase treatment adherence and can be useful for managing medication at home [2].

Another systematic review of medication reminder applications identified hundreds of available apps and found that common useful features include flexible scheduling, medication history tracking, snooze options, and visual aids [3].

However, the same review also showed that many applications provide only a limited number of desirable features, which indicates that there is still a need for more complete and integrated medication management solutions.

For elderly patients, medication reminder technologies can be especially useful because they support independent living and reduce the risk of missed or incorrect doses. Recent systematic reviews indicate that reminder technologies may contribute to positive health outcomes, although more high-quality research is still needed to evaluate their full impact on user experience, cost-effectiveness, and healthcare service utilization.

2.3.3 Health Monitoring and Vital Signs Tracking

Medication management alone is not always enough for elderly patients, especially those with chronic diseases such as hypertension, diabetes, and cardiovascular conditions. These patients often need to monitor vital signs regularly, including blood pressure, blood glucose, weight, and temperature. Recording these measurements over time helps both patients and doctors understand health trends and detect abnormal changes earlier.

Mobile health applications can support this process by allowing users to enter health readings, view historical data, and monitor trends through charts and summaries. Some existing applications, such as MyTherapy, Medisafe, and CareClinic, support tracking of measurements such as blood pressure, blood sugar, weight, and other health indicators.

In the Smart Medication Management System, the health readings module allows patients to record blood pressure, blood sugar, weight, and temperature. The application also provides history screens, statistics, and visual feedback for different reading types. This makes the system more useful than a simple pill reminder because it combines medication adherence with continuous health monitoring.

2.3.4 Cloud-Based Healthcare Systems

Cloud computing plays an important role in modern healthcare applications because it enables secure data storage, synchronization across devices, and remote access to patient information. In mobile healthcare systems, cloud services allow patient records to be stored centrally and accessed by authorized users, such as doctors and caregivers, when needed.

Cloud Firestore supports offline data persistence, which allows applications to cache data locally so that users can read, write, listen to, and query cached data even when the device is offline. When the device reconnects to the internet, local changes are synchronized with the cloud backend. This feature is important for healthcare applications because patients may need access to critical information even during unstable network conditions.

The Smart Medication Management System uses Firebase services, including Firebase Authentication, Cloud Firestore, Firebase Cloud Messaging, and Crashlytics. The project also supports offline persistence and cloud synchronization, allowing users to continue using important parts of the application when internet connectivity is temporarily unavailable.

2.3.5 Security and Privacy in Mobile Healthcare Applications

Healthcare applications handle sensitive personal and medical data, so security and privacy are critical requirements. Patient data must be protected from unauthorized access, modification, leakage, and misuse. Common security mechanisms include authentication, authorization, encrypted storage, secure communication, audit logging, input validation, and access control.

Firebase Authentication supports user authentication and session persistence, while Cloud Firestore security rules can be used to restrict database access based on authenticated users and user roles. This is particularly important in applications that include different user types, such as patients and doctors.

The Smart Medication Management System applies security mechanisms such as user authentication, email verification, role-based access control, Firestore security rules, input sanitization, rate limiting, and audit logs. It also includes device integrity checks to detect rooted or jailbroken devices and restrict sensitive caching when the device is considered unsafe.

2.4 Existing Systems Comparison

Several mobile applications already exist for medication reminders and health tracking. However, each system has different strengths and limitations. This section reviews selected existing systems and compares them with the proposed **Smart Medication Management System for Elderly People**.

2.4.1 MyTherapy

MyTherapy is a medication reminder and health tracker application. It provides medication reminders, intake documentation, refill reminders, health measurements tracking, symptom tracking, and health reports that can be shared with doctors. It also supports measurements such as blood sugar, blood pressure, and weight.

Although MyTherapy provides a strong set of medication and health tracking features, the Smart Medication Management System focuses more specifically on elderly users, bilingual Arabic-English support, QR-code-based doctor access, and role-based patient-doctor interaction within the same application.

2.4.2 Medisafe

Medisafe is a well-known medication management application that supports medication reminders, complicated dosing schedules, measurement tracking, caregiver support through Medfriend, drug interaction warnings, refill reminders, and appointment reminders. It also allows users to share medication reports with healthcare providers.

Medisafe is strong in medication adherence and caregiver support, but the Smart Medication Management System is designed with a localized academic focus on elderly patients, Arabic and English interfaces, QR-code-based medical data sharing, and secure role-based access between patients and doctors.

2.4.3 CareClinic

CareClinic is a personal health tracking application that supports medication tracking, symptom tracking, health records, health journals, blood pressure tracking, blood sugar tracking, mood tracking, and sharing reports with doctors and caregivers. It also supports integration with health platforms and wearable devices.

2.4.4 Comparison Table

Table 1: Comparison between Existing Medication Management Systems and the Smart Medication Management System

Feature	MyTherapy	Medisafe	CareClinic	Dosecast	Smart Medication Management System
Medication reminders	Yes	Yes	Yes	Yes	Yes
Dose tracking	Yes	Yes	Yes	Yes	Yes
Health readings tracking	Yes	Yes	Yes	Limited	Yes
Doctor report sharing	Yes	Yes	Yes	Yes	Yes
QR-code-based sharing	Limited	Not a main highlighted feature	Not a main highlighted feature	Not a main highlighted feature	Yes
Arabic-English support	Not the main focus	Not the main focus	Not the main focus	Not the main focus	Yes
Patient and doctor roles	Limited	Limited / caregiver-oriented	Limited / caregiver-oriented	Limited	Yes
Offline support	Limited / not clearly emphasized	Limited / not clearly emphasized	Limited / not clearly emphasized	Yes for reminders	Yes
Elderly-oriented UI	General user interface	General user interface	General user interface	General medication interface	Yes
Security and access control	Privacy and security mentioned	Privacy and compliance mentioned	Encryption and privacy mentioned	Encryption in transit mentioned	Authentication, RBAC, Firestore Rules, audit logs, and rate limiting

Table 1 Comparison Table

2.5 Technologies Review

2.5.1 Flutter

Flutter is an open-source framework developed by Google for building natively compiled, multi-platform applications from a single codebase. It supports mobile, web, desktop, and embedded platforms, and it allows developers to create consistent user interfaces across different devices.

Flutter is suitable for this project because the system requires a modern mobile interface, fast development, reusable UI components, and the possibility of future expansion to iOS or other platforms. In addition, Flutter's widget-based architecture supports the creation of responsive and accessible interfaces, which is important for elderly users.

2.5.2 Firebase

Firebase is used as the backend infrastructure for the Smart Medication Management System. It provides services such as authentication, cloud database storage, push notifications, analytics, and crash reporting. These services reduce the need to build a complete backend server from scratch and allow faster development of secure and scalable mobile applications.

In this project, Firebase Authentication is used for user login and registration, Cloud Firestore is used for storing medication and health data, Firebase Cloud Messaging supports notifications, and Crashlytics helps track application crashes and improve reliability.

2.5.3 Cloud Firestore

Cloud Firestore is a cloud-hosted NoSQL database that supports real-time synchronization and offline persistence. Offline persistence allows the app to cache data locally, continue reading and writing data while offline, and synchronize changes when the device reconnects to the internet.

This technology is highly suitable for the proposed healthcare application because elderly patients may need access to medication schedules and recent health records even if the internet connection is unstable. The project uses Firestore together with security rules to protect patient records and enforce role-based access control.

2.5.4 Notification Technologies

Medication reminders depend heavily on reliable notification technologies. Local notifications are useful for scheduled reminders on the device, while cloud messaging can be used for remote alerts and communication. Existing medication reminder apps commonly include flexible reminders, snooze features, dose history, and refill alerts because these features help patients follow complex medication schedules.

The Smart Medication Management System implements an advanced notification system that allows patients to interact directly with reminders using actions such as **Mark Taken**, **Skip**, and **Snooze**. It also handles pending actions when the application is opened again after being closed, which improves medication adherence tracking.

2.5.5 QR Code Technology

QR codes can be used in healthcare applications to quickly share information, identify patients, or link users to digital records. In the Smart Medication Management System, QR-code-based sharing allows a doctor to access a patient's medical information quickly and in an organized way, after proper authorization. This improves the efficiency of clinical follow-up and reduces the need for manual data transfer.

2.6 Research Gap

Based on the literature review and existing systems comparison, several research gaps can be identified in current medication management and mobile healthcare applications.

First, many existing medication reminder applications focus mainly on basic reminder functions and dose tracking. Although these features are useful, they are not always enough for elderly patients who may need a more complete healthcare support system. Elderly patients often require medication reminders, health readings tracking, secure data storage, doctor access, and simple communication features within one integrated platform. However, many existing applications provide these features separately or in a limited form.

Second, many available systems are designed for general users rather than elderly patients. Elderly users usually need simple navigation, clear visual elements, readable text, and easy access to the most important functions. Complex interfaces may reduce usability and make it harder for elderly patients to manage their medications and health records effectively. Therefore, there is a need for a system that focuses on elderly-friendly design and reduces the number of steps required to complete essential tasks.

Third, most global medication management applications are not specifically designed for Arabic-speaking users. Language support is an important factor in healthcare applications because users need to understand medication instructions, reminders, health readings, and system messages clearly. Limited Arabic support may reduce the effectiveness of the application for local users. Therefore, bilingual Arabic-English support represents an important gap that should be addressed in the Smart Medication Management System.

Finally, security and privacy remain critical challenges in healthcare applications because the system handles sensitive medical and personal data. Applications must protect patient information from unauthorized access, misuse, and data leakage. This requires authentication, role-based access control, secure cloud storage, and controlled data sharing between patients and doctors.

Therefore, the proposed Smart Medication Management System for Elderly People addresses these gaps by combining medication reminders, health readings tracking, QR-code-based doctor access, bilingual Arabic-English support, offline support, cloud synchronization, and role-based security in one elderly-friendly mobile healthcare platform.

Chapter 3 Planning and Analysis

3.1 Planning

The project follows an **Agile Iterative Development** approach. This methodology includes **sprint planning**, **daily progress through regular code commits and feature implementation**, **continuous testing on multiple devices and screen sizes**, **code review and refactoring for quality assurance**, and **incremental feature releases with version control**. The document states that this methodology ensures steady progress, early bug detection, and continuous improvement throughout the development lifecycle.

The project is also supported by a **layered architecture pattern** that includes the **Presentation Layer**, **Business Logic Layer**, **Service Layer**, **Data Layer**, and **Infrastructure Layer**. According to the book, this architecture ensures **testability**, **maintainability**, **scalability**, and **reusability**, which reflects an organized planning structure for both implementation and future enhancement.

3.1.1 Feasibility Study and Estimated Cost

The closest related content appears under **Tools and Technologies Used**, where the system is described as being built with **Flutter**, **Dart**, **Firebase Authentication**, **Cloud Firestore**, **Firebase Cloud Messaging**, **Provider**, and additional packages for **localization**, **notifications**, and **secure storage**. The document explains that this technology stack was selected to ensure **reliability**, **performance**, **maintainability**, **security**, and **scalability**.

3.2 Functional Requirements

The Smart Medication Management System for Elderly People provides comprehensive functionality organized into distinct modules:

3.2.1 User Access Requirements

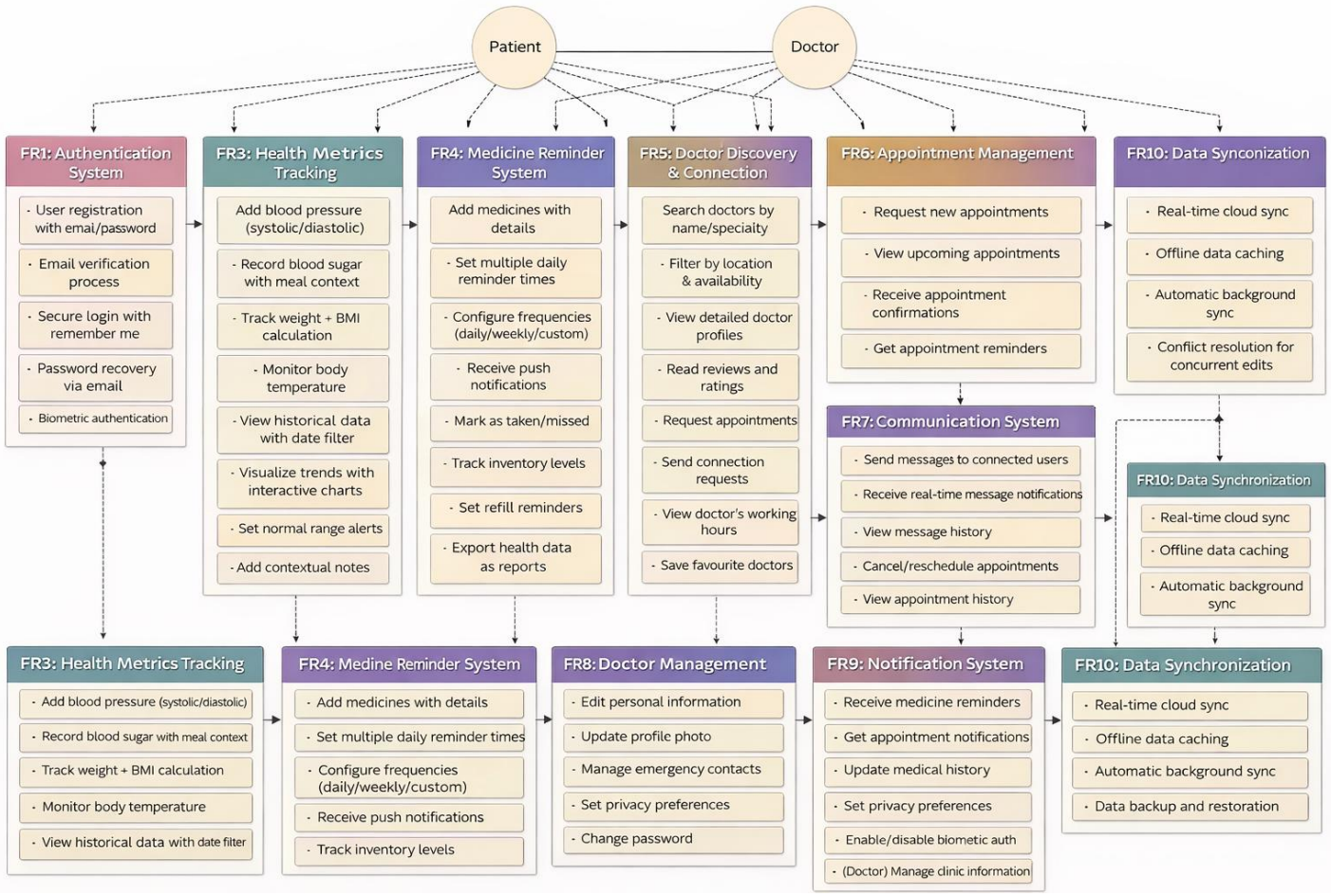


Figure 4 User Access Requirements

3.3 Non-functional Requirements

3.3.1 Properties of the Application

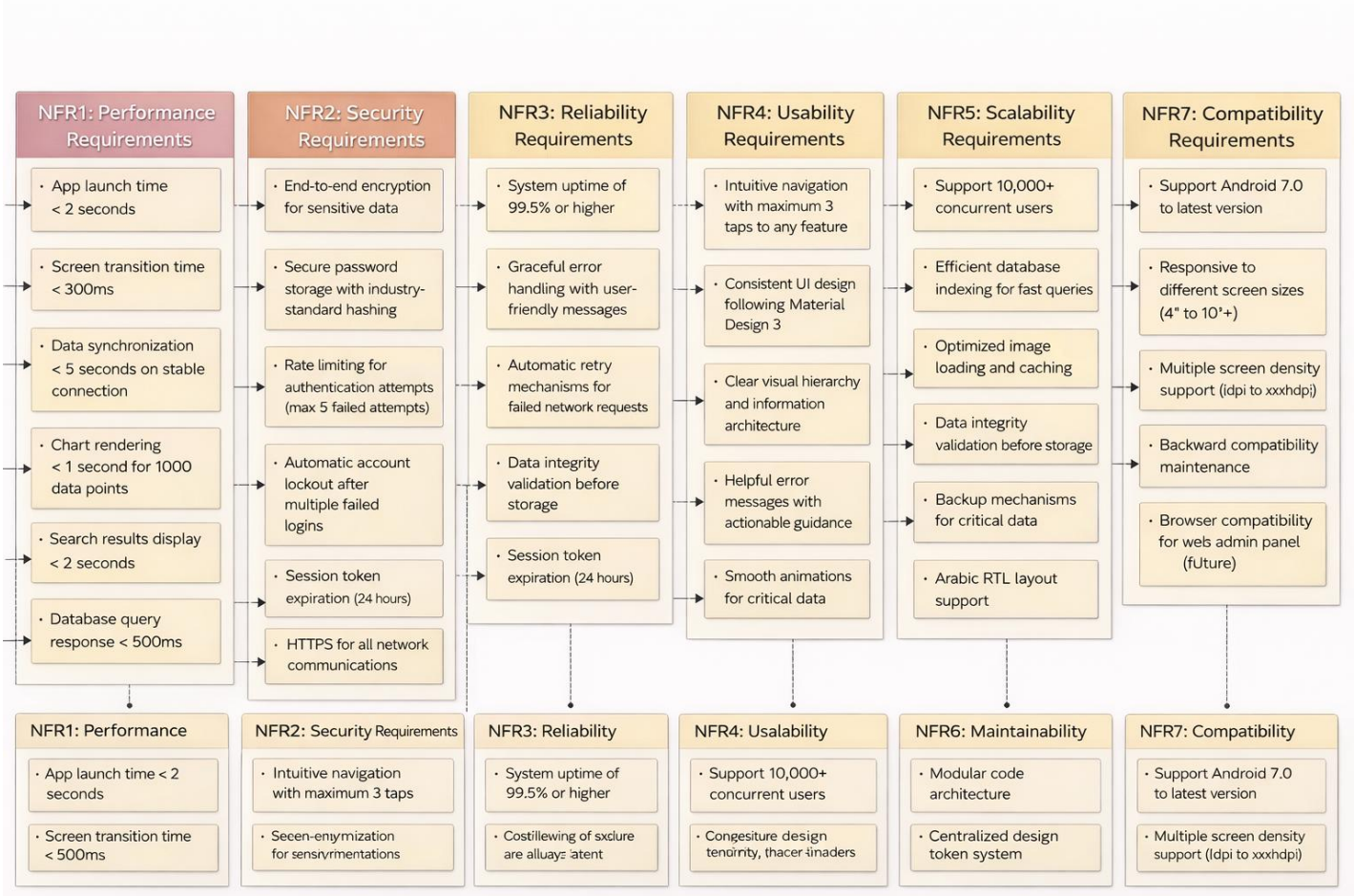


Figure 5 Non-functional Requirements

3.3.2 Permissions Needed

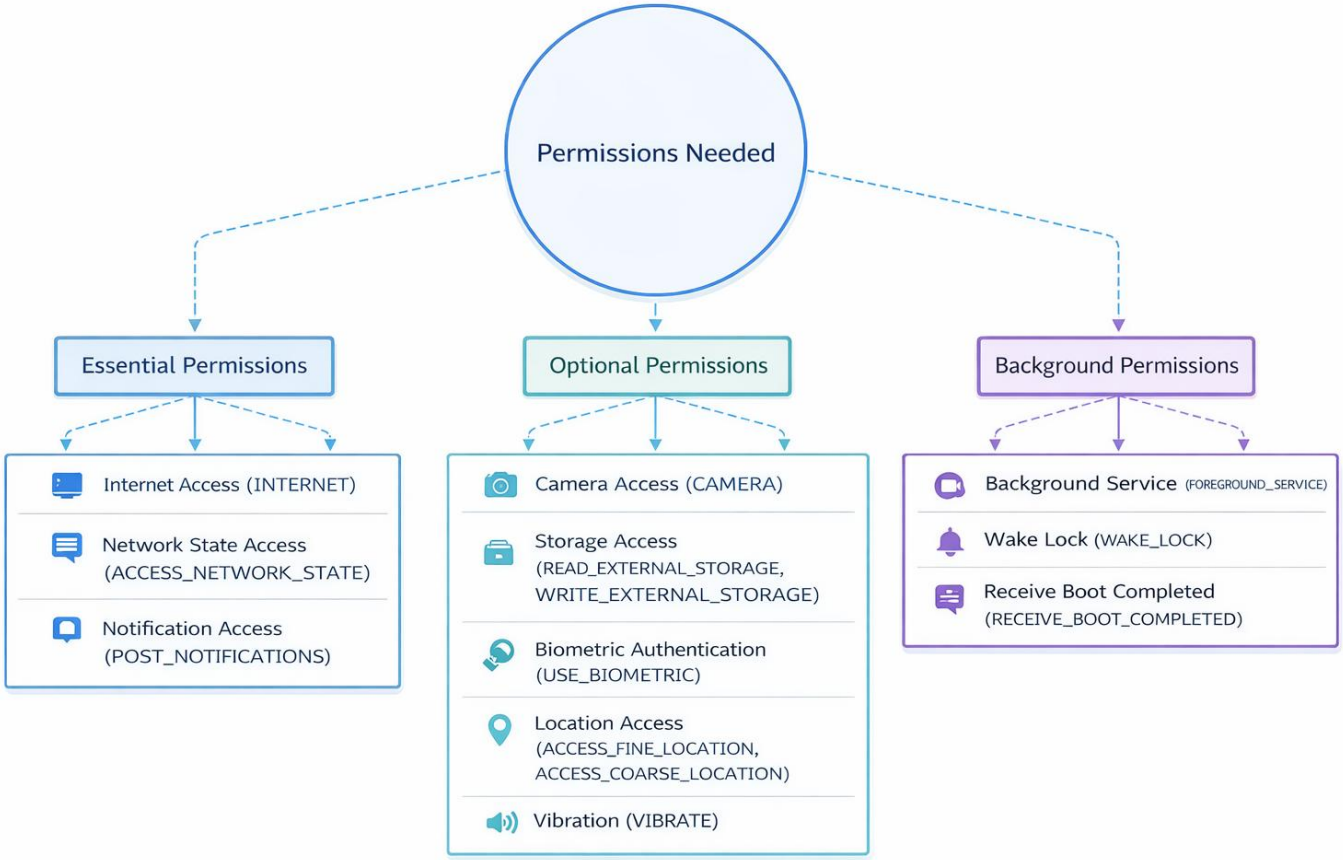


Figure 6 Permissions Needed

3.4 UML Diagrams

3.4.1 Context Diagram

The Smart Medication Management System for Elderly People operates within a comprehensive ecosystem connecting patients, doctors, and external services.

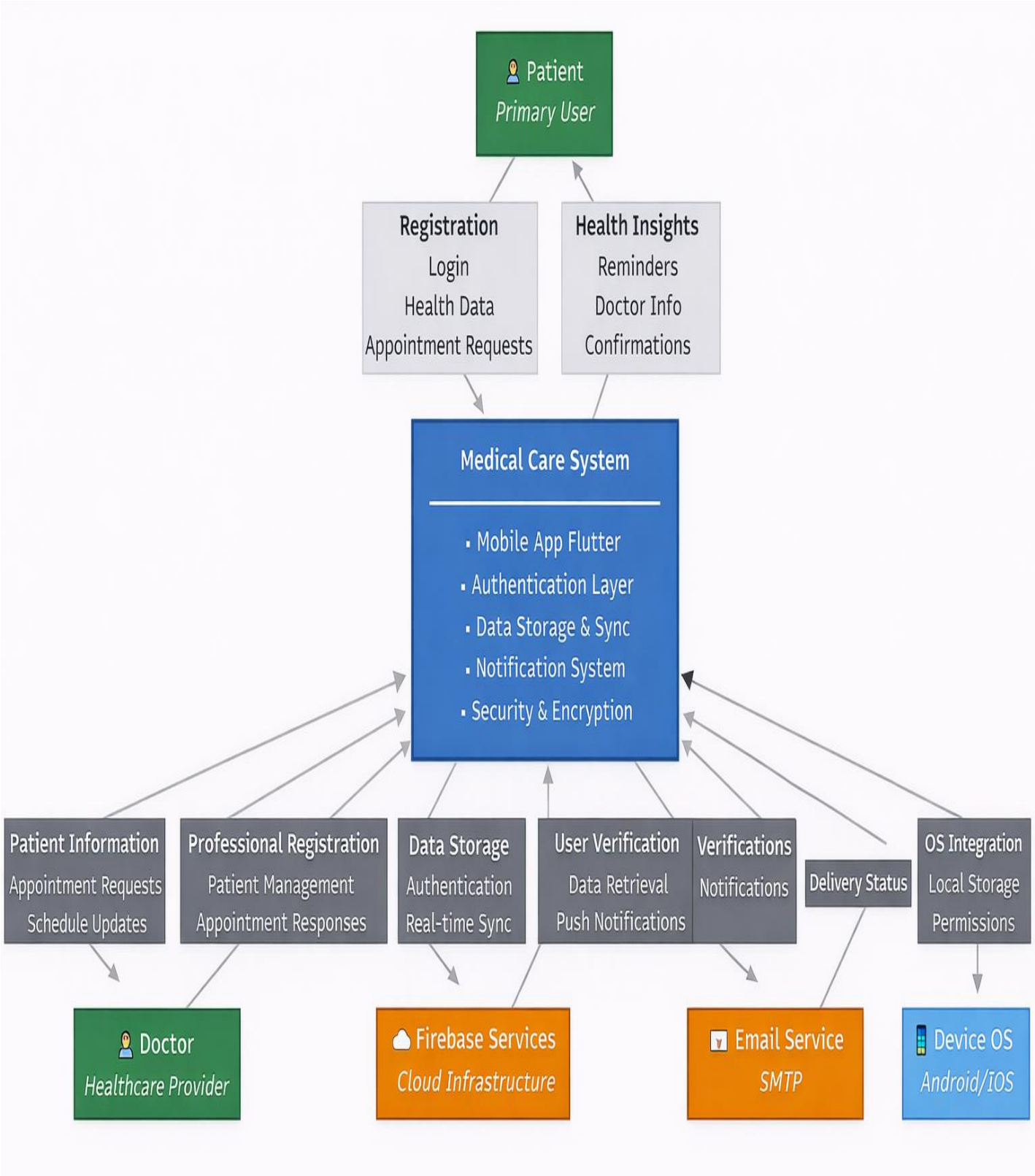


Figure 7 Context Diagram

3.4.2 Data Flow Diagram

The Data Flow Diagram (DFD) represents how data moves through the Smart Medication Management System for Elderly People system, showing the major processes, data stores, and data transformations.

Level 0 DFD (Context Level) - Main Data Flows:

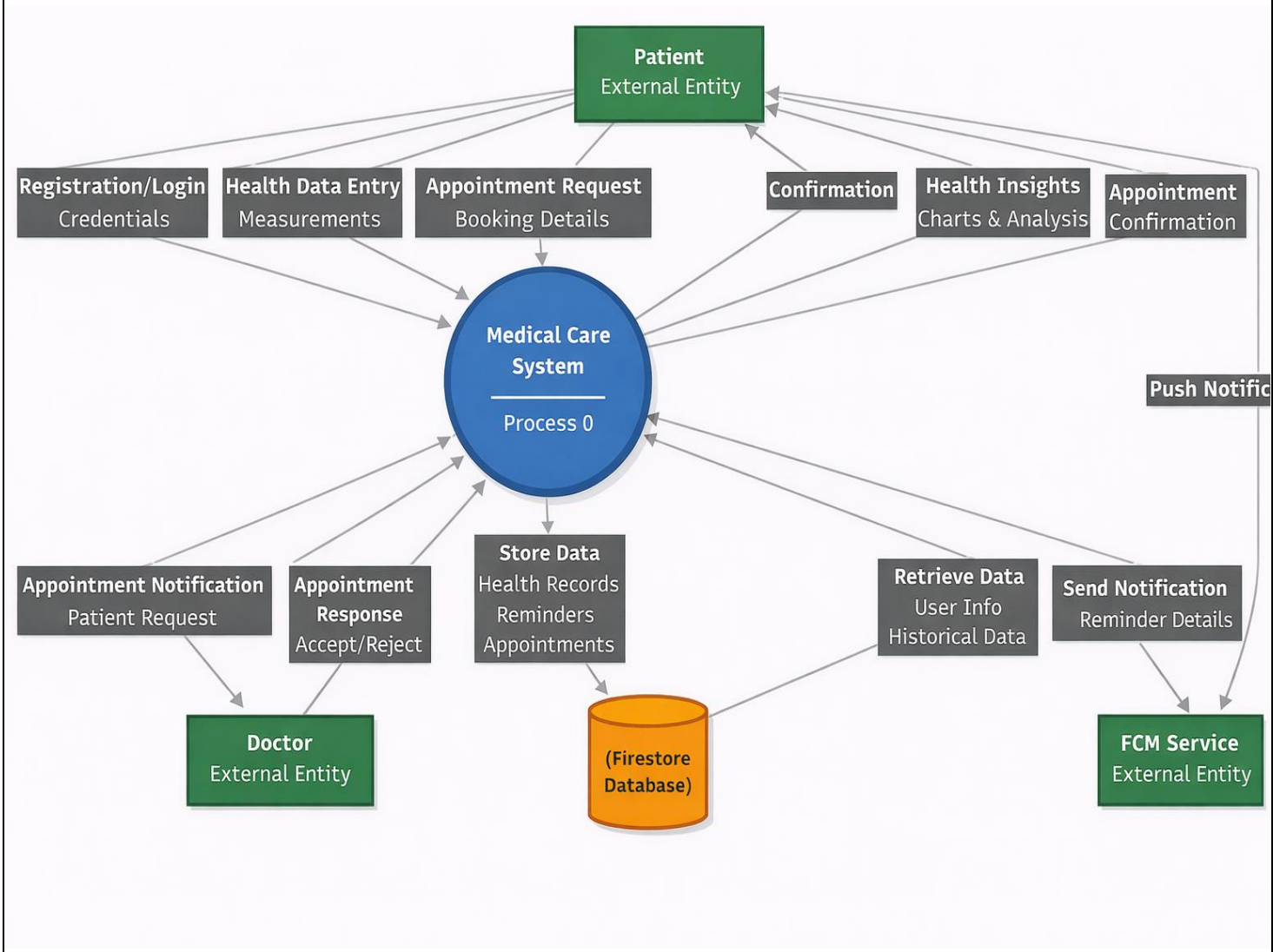


Figure 8 - Level 0 Data Flow Diagram

Level 1 DFD - Major Processes:

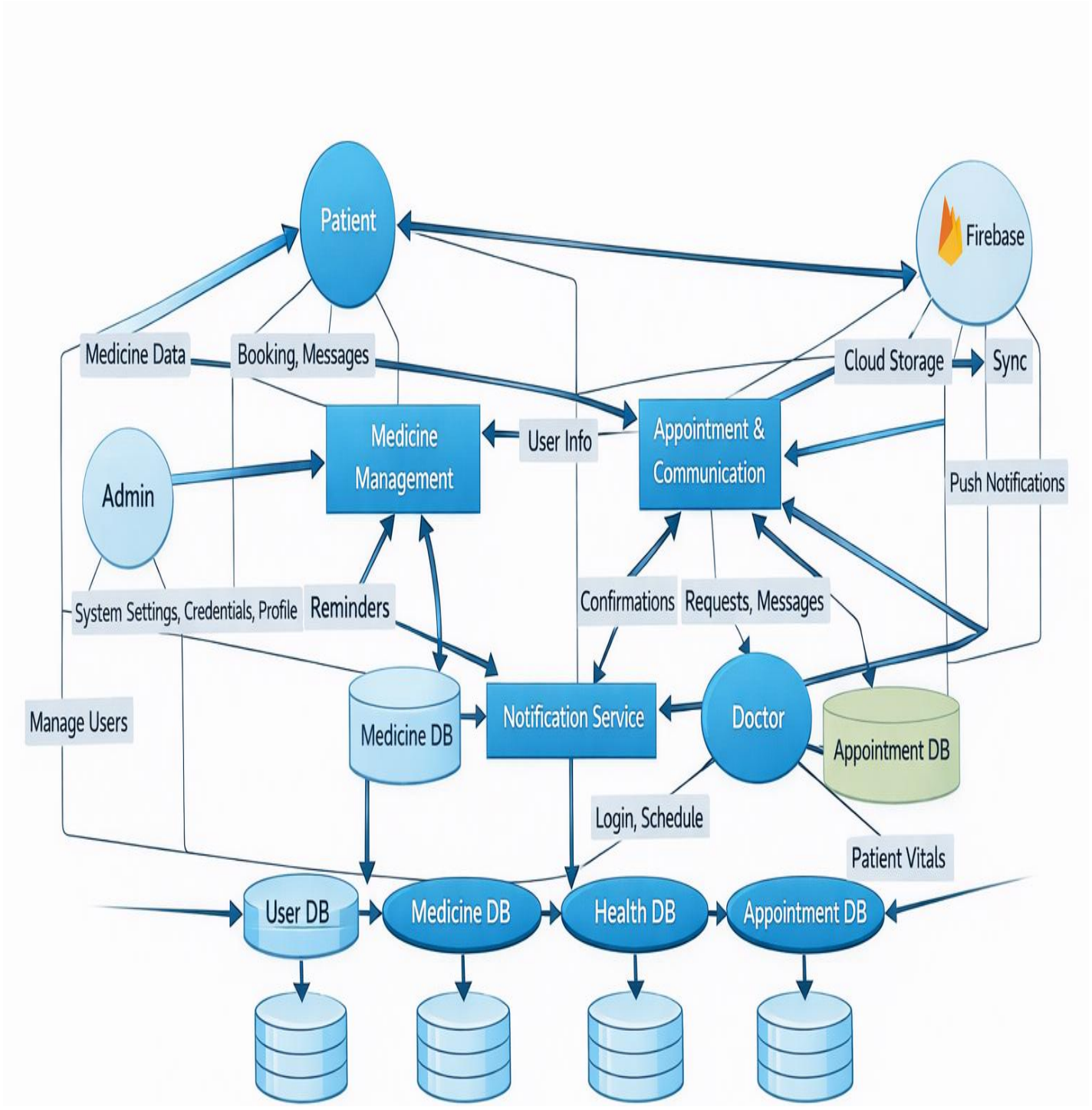


Figure 9 DFD - Level 1 Data Flow Diagram - Major Processes

3.4.3 **Activity Diagram**

Activity diagrams show the workflow of stepwise activities and actions with support for choice, iteration, and concurrency. Below are key activity flows in the Smart Medication Management System for Elderly People:

Activity 1: Patient Registration Flow

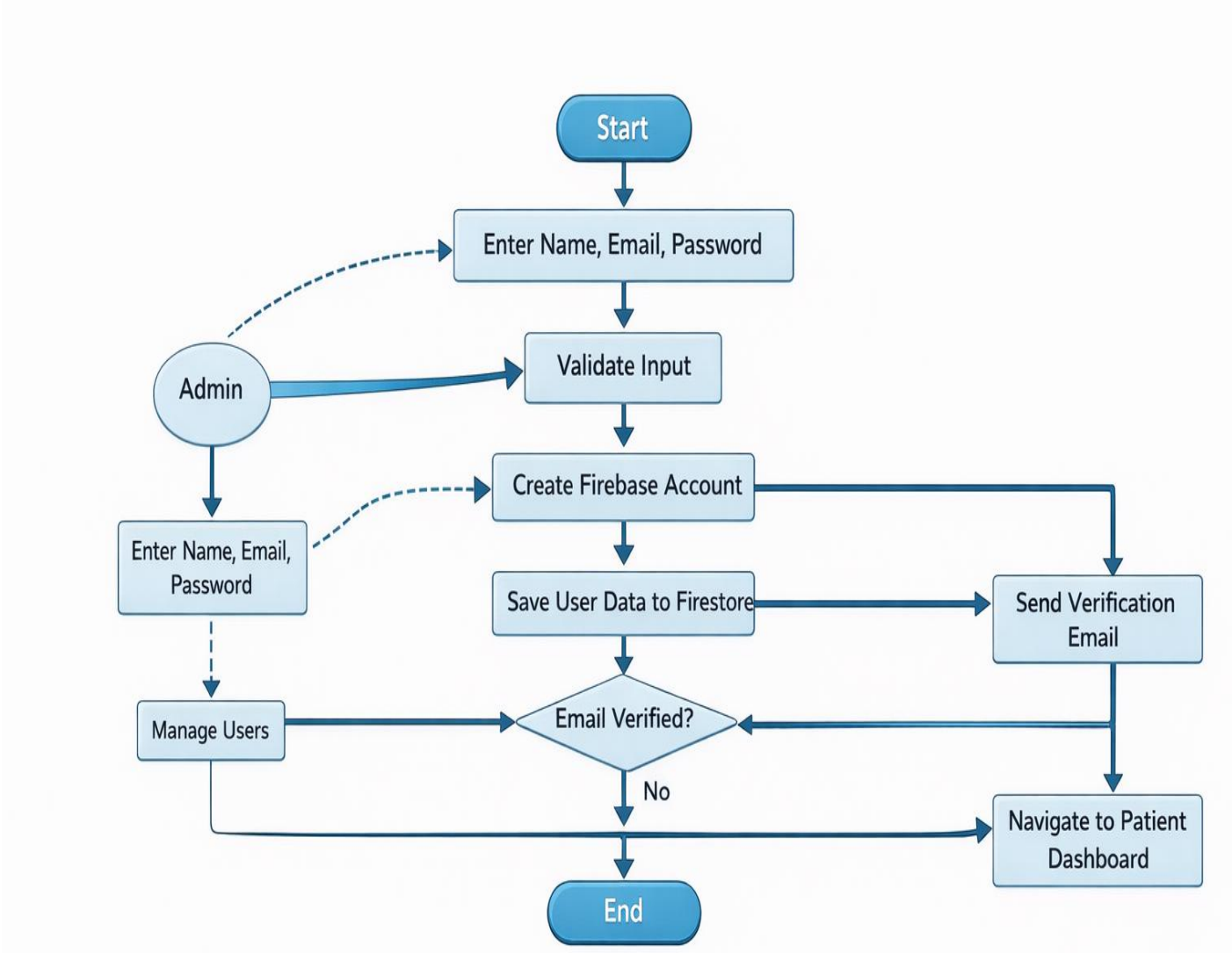


Figure 10 Patient Registration Flow

Activity 2: Recording Health Metric

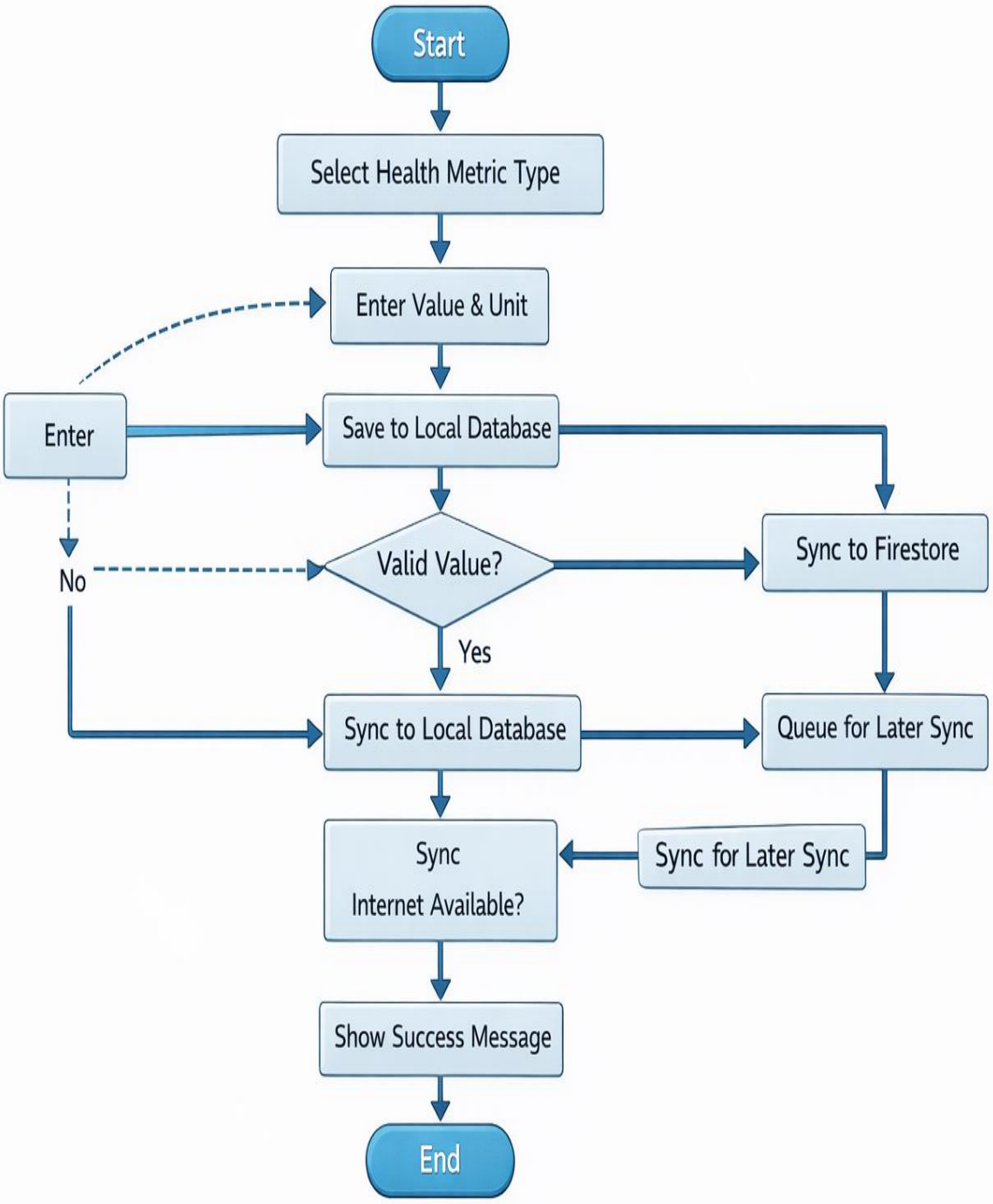


Figure 11 Activity 2: Recording Health Metric

Activity 3: Setting Medicine Reminder

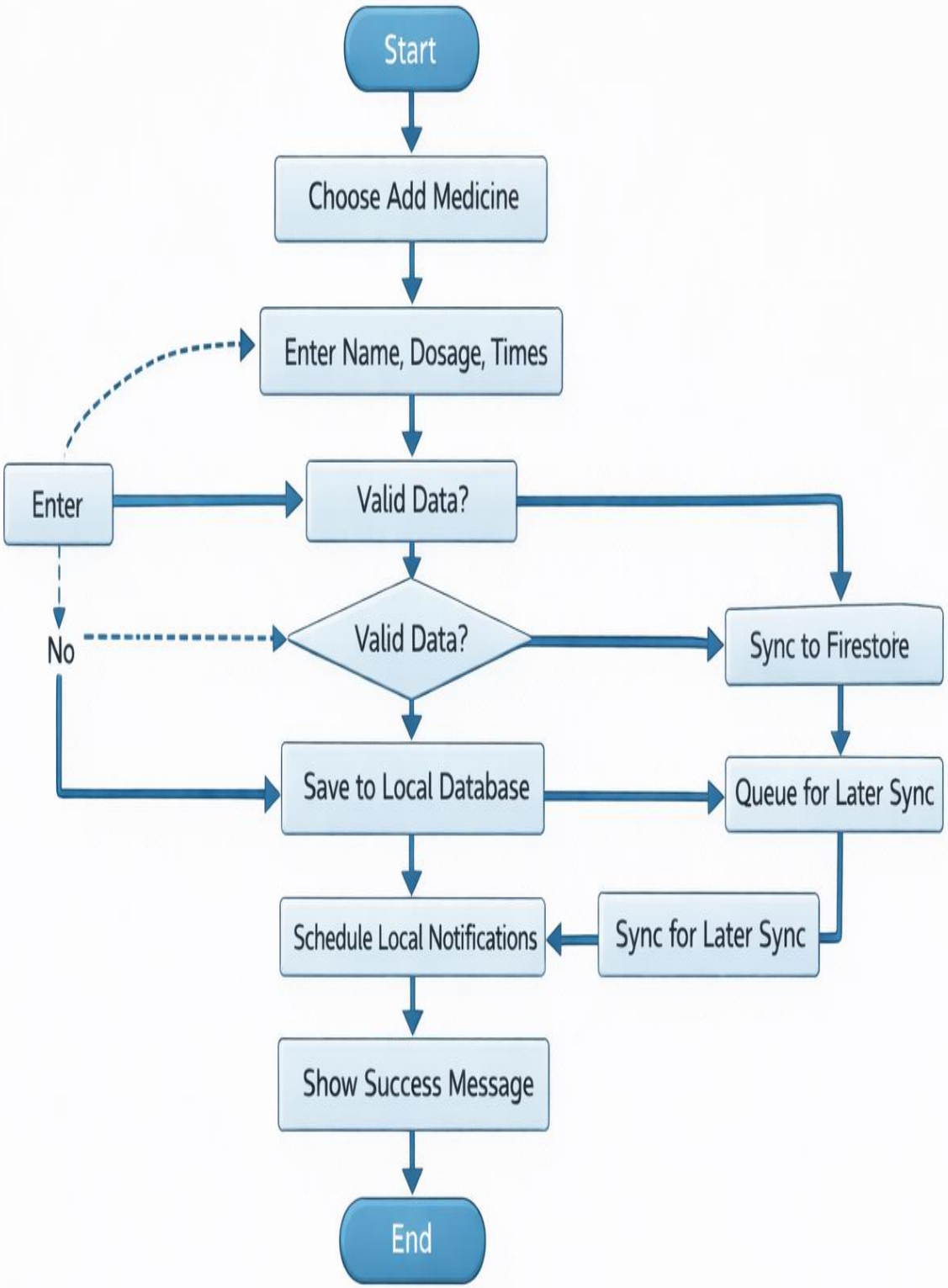


Figure 12 Activity Diagram

3.4.4 Flow Chart Diagram

Flow charts provide detailed step-by-step visualization of specific processes with decision points and alternative paths:

Flowchart 1: User Login Process

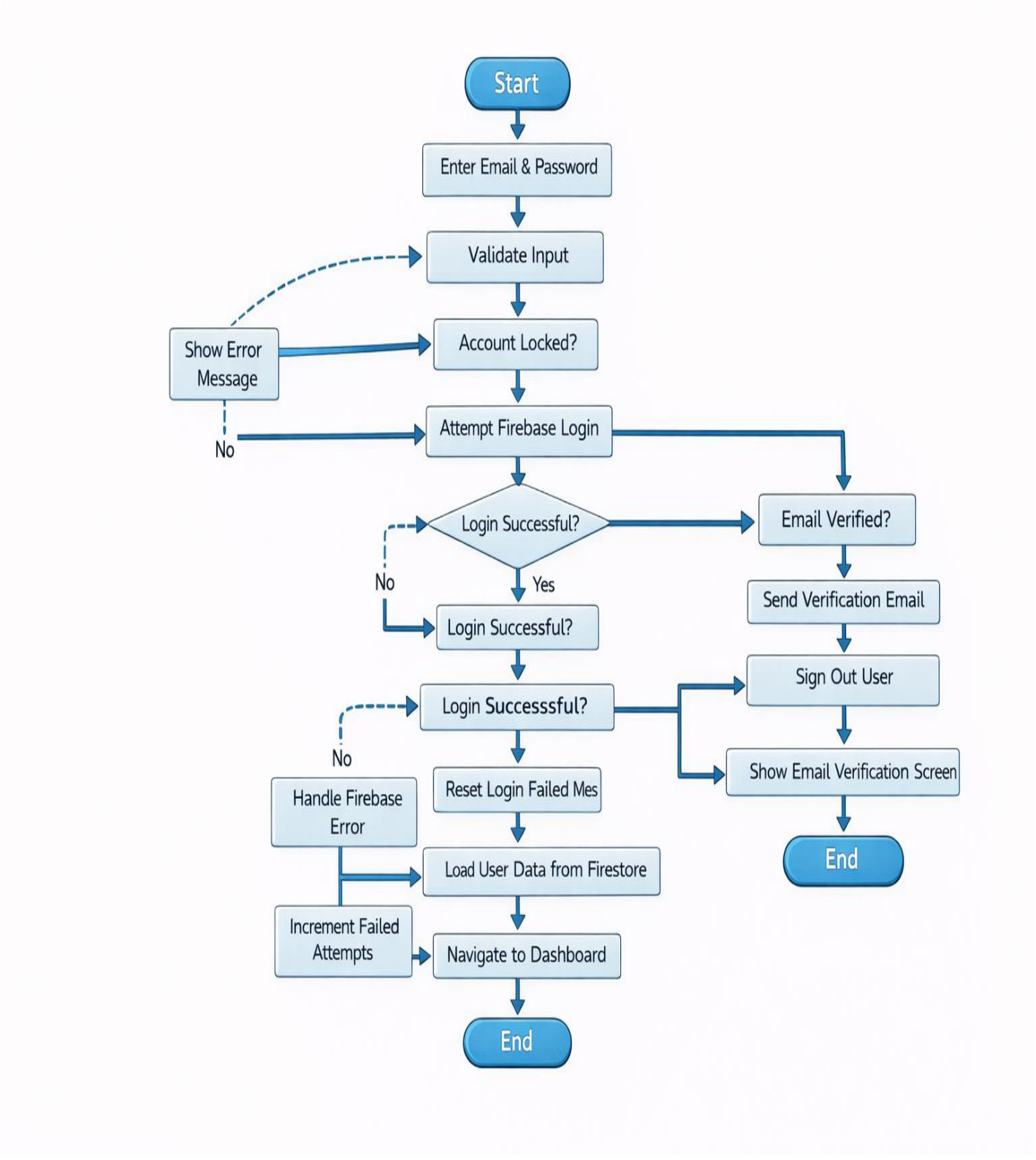


Figure 13 Flowchart Diagram / User Login Process

Flowchart 2: Medicine Reminder Notification (Background Service)

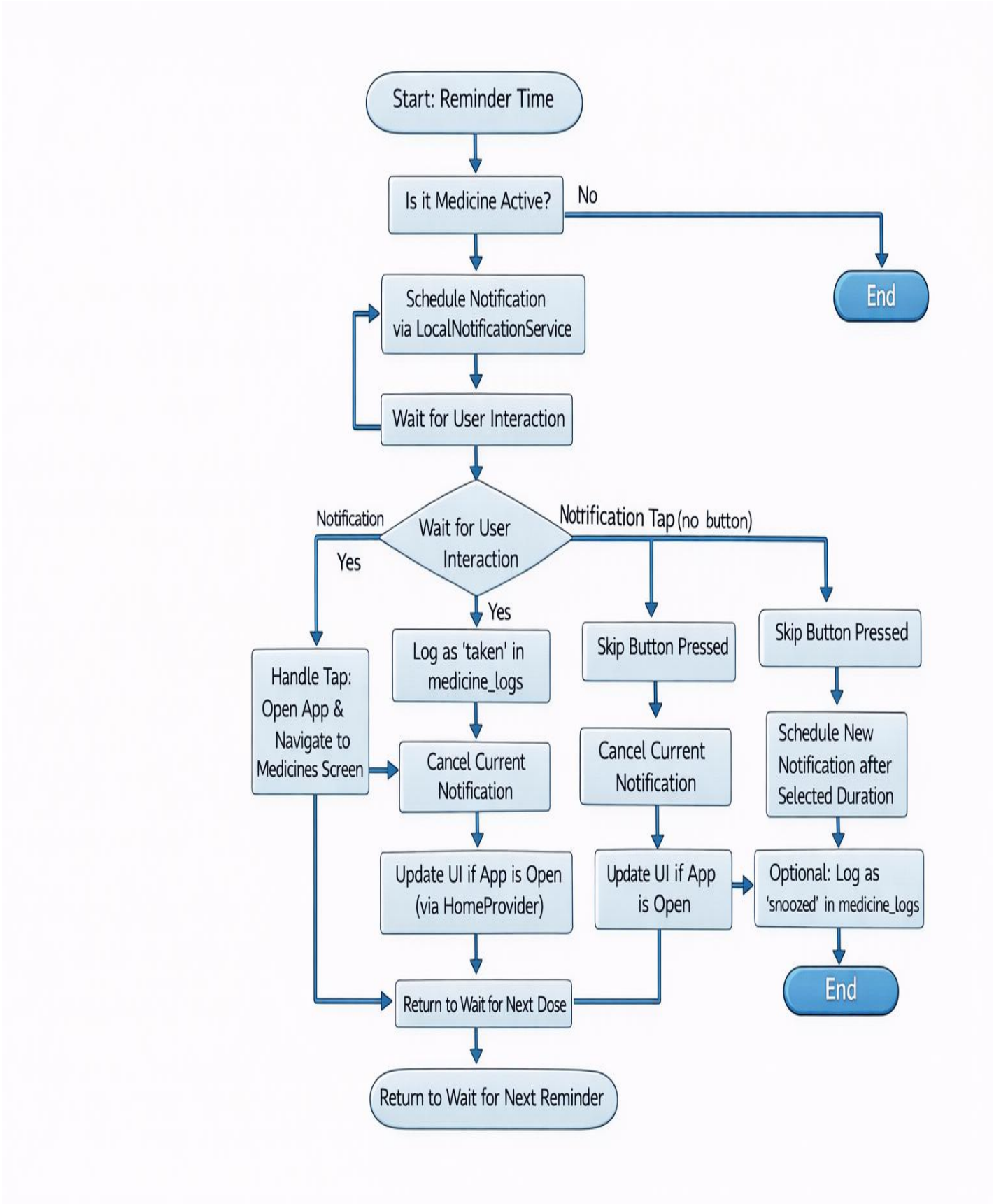


Figure 14 Flowchart Medicine Reminder

Flowchart 3: Health Data Validation and Storage

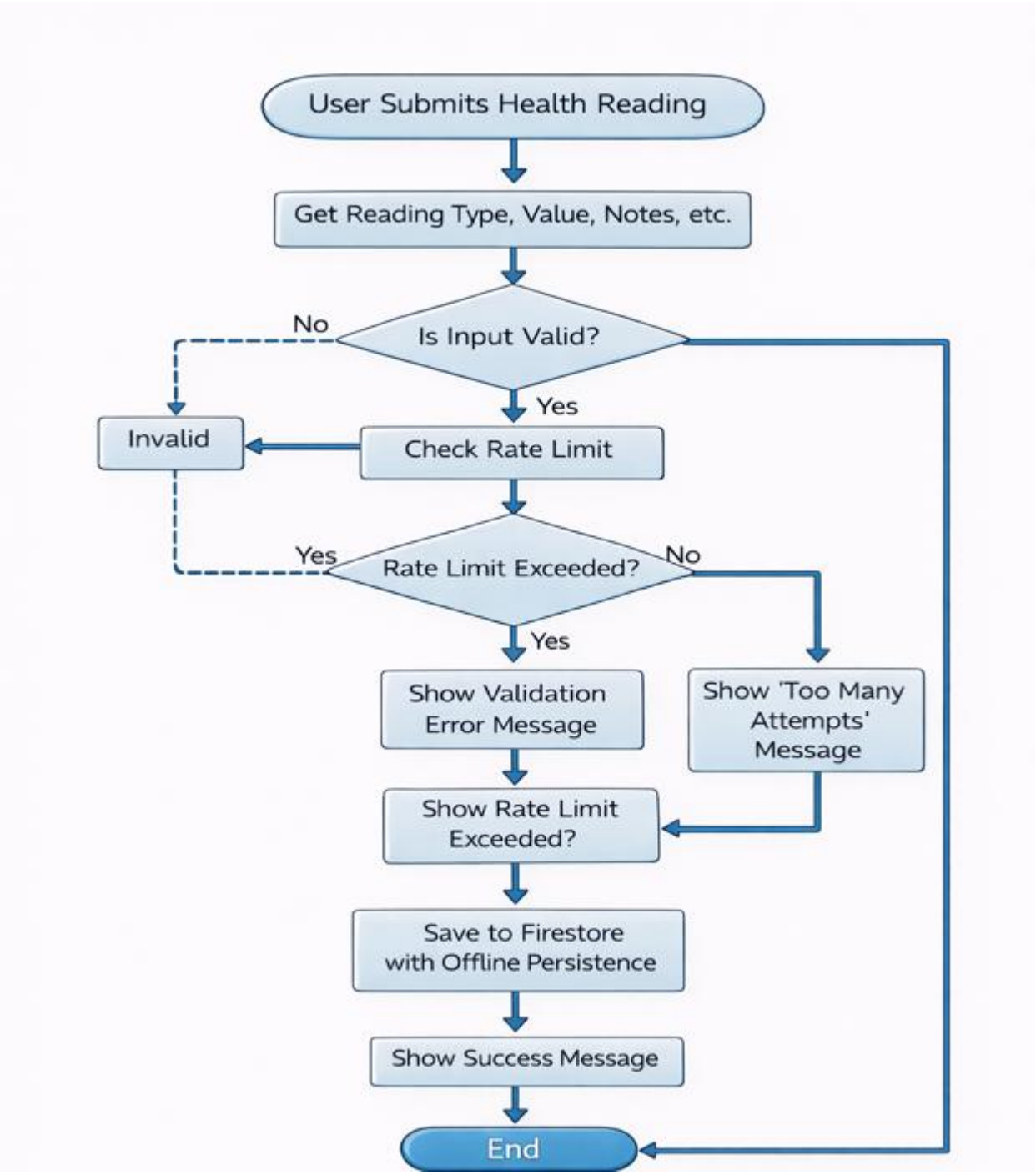


Figure 15 Flowchart - Health Data Validation

3.4.5 Sequence Diagram

The sequence diagram illustrates the login and authentication workflow of the Smart Medication Management System for Elderly People. For better readability, the diagram is divided into three phases: initialization, authentication method selection, and login execution. These phases show how the system checks stored credentials, biometric availability, account lock status, Firebase Authentication, email verification, Firestore user data, and failed login attempts.

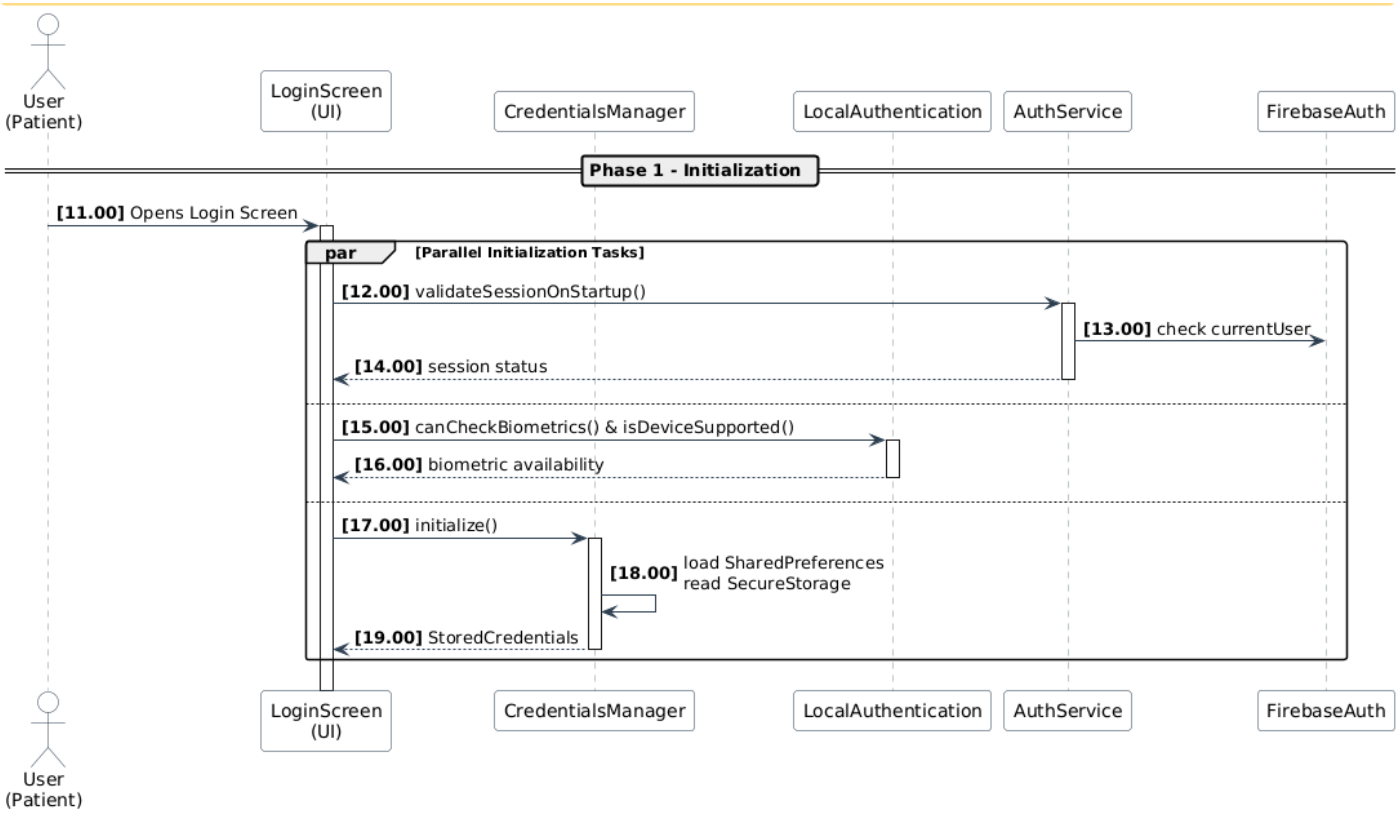


Figure 16 Login Authentication Sequence Diagram - Phase 1 Initialization

This phase describes the initialization process that occurs when the patient opens the login screen. The system validates the current session, checks biometric availability, and loads any stored credentials from secure local storage before continuing to authentication.

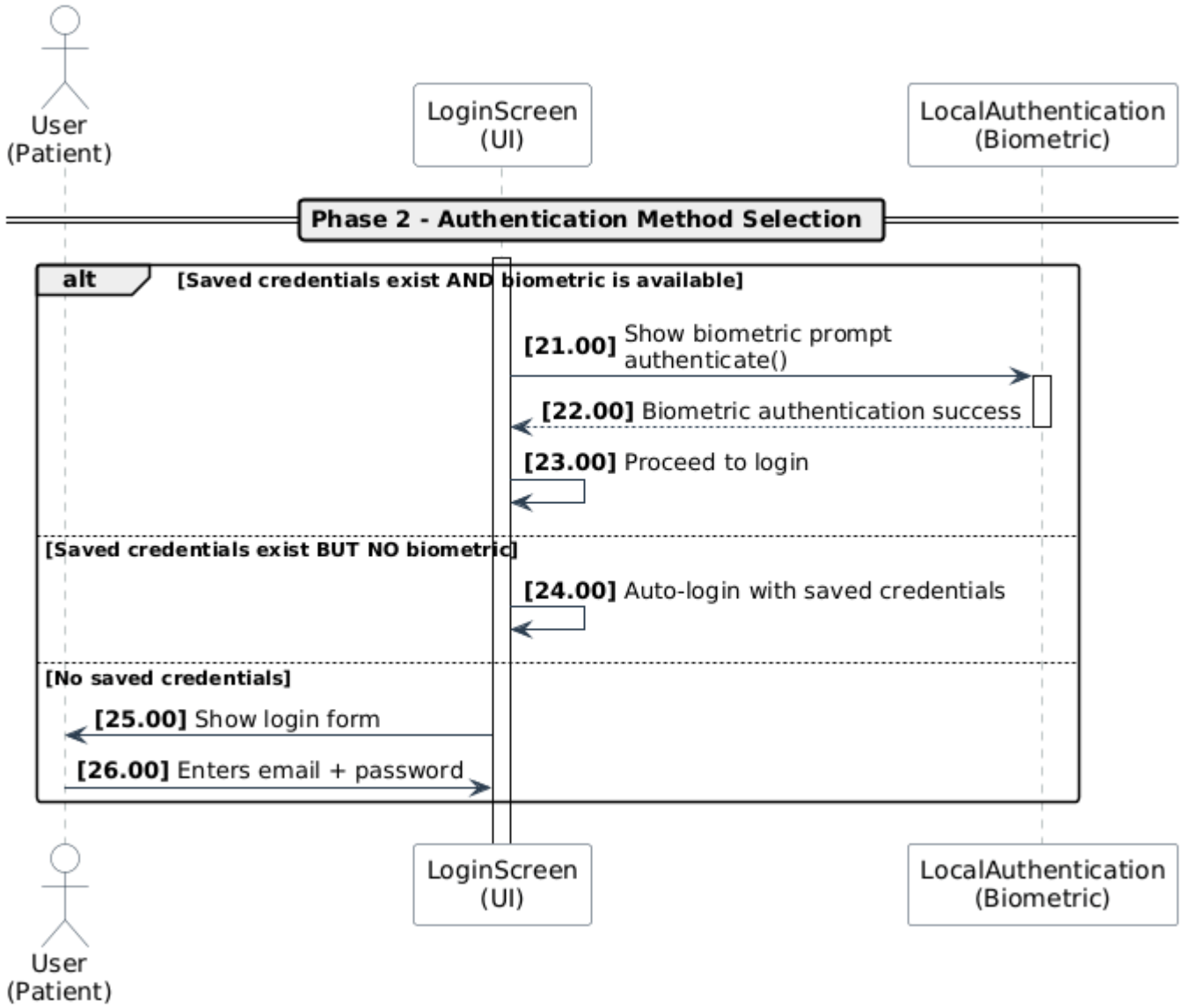


Figure 17 Login Authentication Sequence Diagram - Phase 2 Authentication Method Selection

This phase explains how the system selects the suitable authentication method. Depending on the availability of saved credentials and biometric support, the user may log in using biometrics, saved credentials, or manual email and password entry.

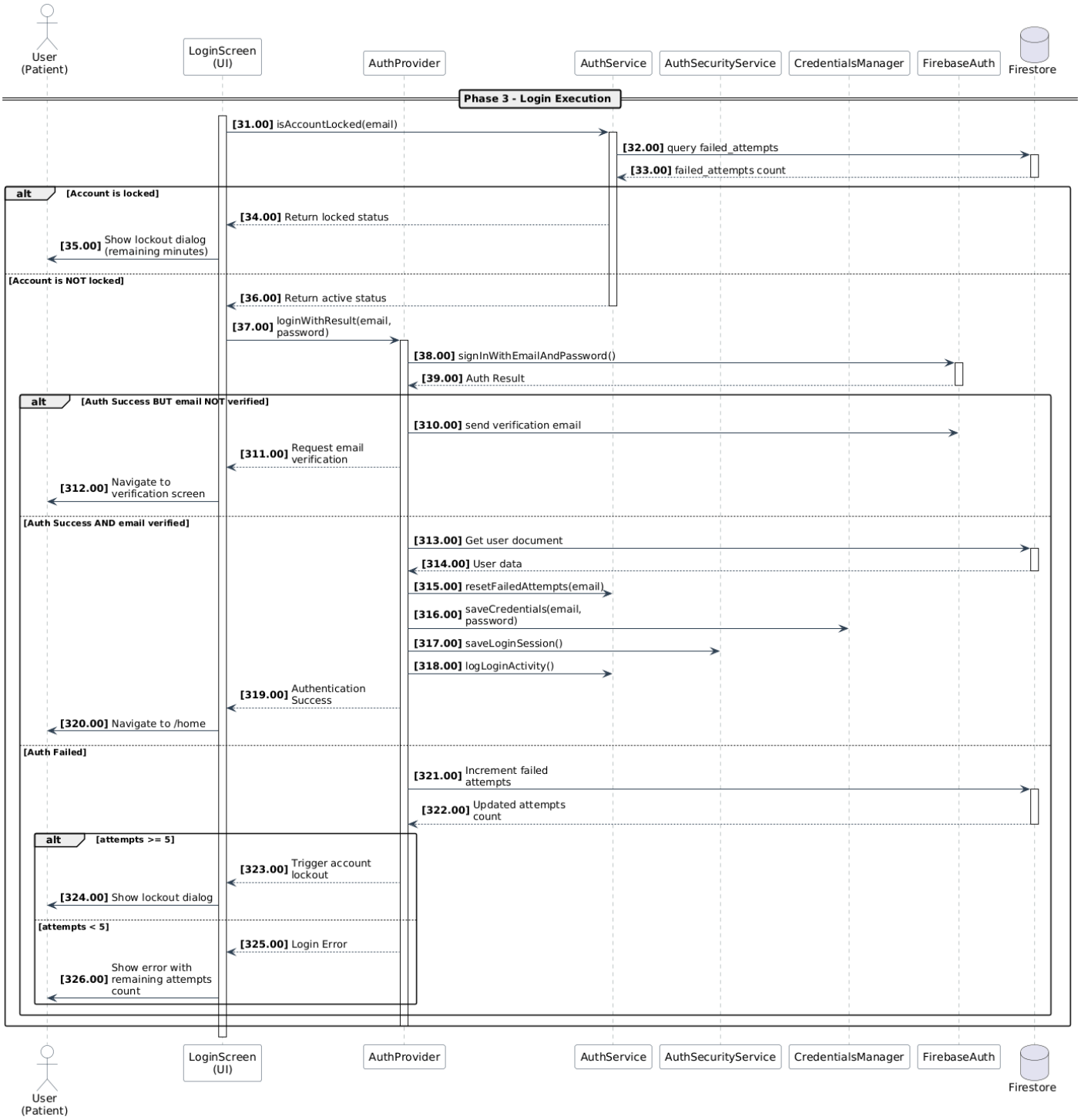


Figure 18 Login Authentication Sequence Diagram - Phase 3 Login Execution

This phase presents the main login execution process. The system checks account lock status, authenticates the user through Firebase Authentication, verifies email status, retrieves user data from Firestore, and handles failed login attempts securely.

3.4.6 Class Diagram

The class diagram shows the static structure of the Smart Medication Management System for Elderly People, illustrating the system's classes, their attributes, methods, and relationships:

Core Classes and Relationships:

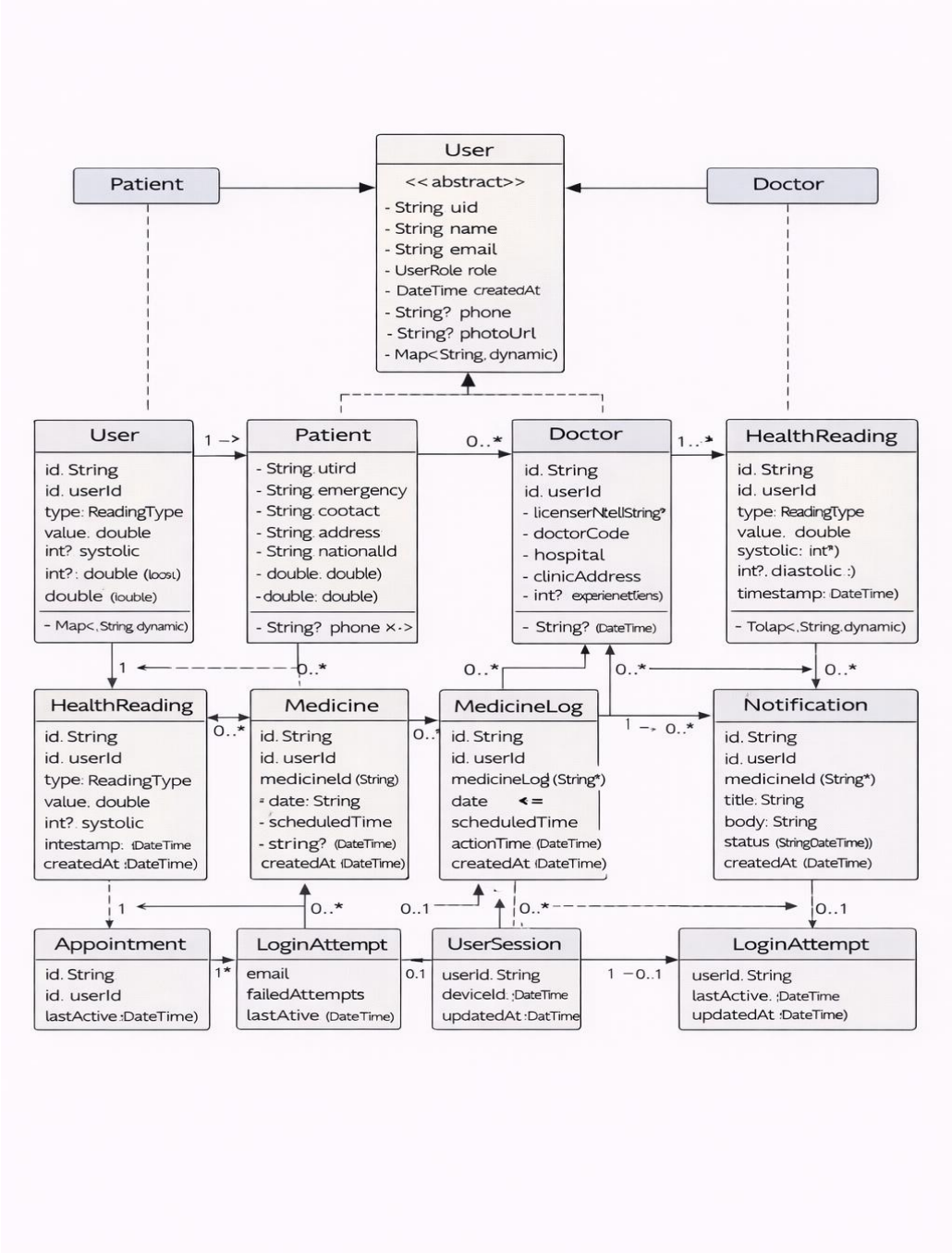


Figure 19 Core Classes and Relationships

Chapter 4: UI/UX Design

4.1 Introduction to UI/UX Design

This chapter presents the comprehensive User Interface (UI) and User Experience (UX) design of the Smart Medication Management System for Elderly People. The design philosophy follows Material Design 3 principles, ensuring a modern, intuitive, and accessible interface for both patients and healthcare providers.

4.2 Splash Screen



Figure 20 Splash Screen \ Loding Screen

Purpose:

The splash screen appears immediately after app launch. It lasts 2–3 seconds while background tasks (checking authentication, loading local data) are performed.

Layout & Function:

- A gradient background (medical blue) with the app logo, name, and tagline.
- A circular progress indicator at the bottom.

4.3 Login Screen

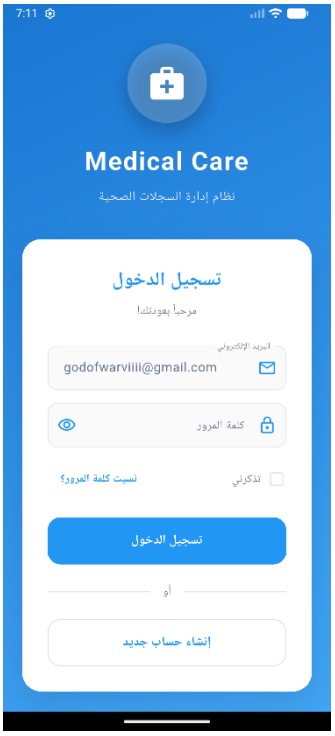


Figure 21 Login Screen

Purpose:

Primary entry point for returning users. Provides secure authentication via email/password, with optional biometric login and "Remember Me".

Key Elements & Actions:

- **Email field** – validates format in real time.
- **Password field** – toggles visibility, checks minimum length.
- **Remember Me checkbox** – when checked, credentials are stored securely (CredentialsManager).
- **Forgot Password link** – opens a dialog to send a reset email.
- **Login button** – disabled until form is valid; shows a loading spinner during authentication.
- **Biometric button** – appears only if device supports biometrics and saved credentials exist. Tapping triggers fingerprint/face authentication.
- **Sign Up link** – navigates to the registration screen.

Security feedback:

- After three failed attempts, a warning dialog appears.

- After five failed attempts, the account is temporarily locked for 15 minutes (progressive lockout)

4.4 Patient Registration Screen

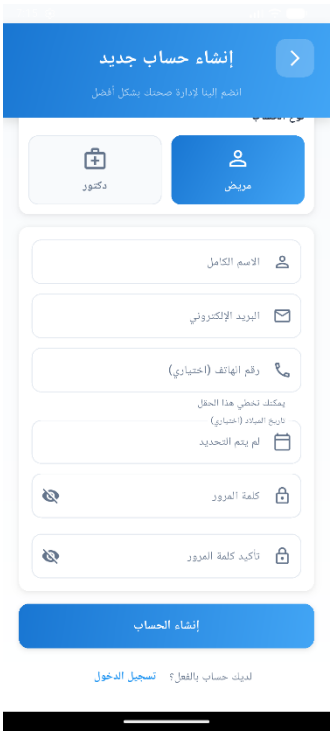


Figure 23 Patient Registration Screen

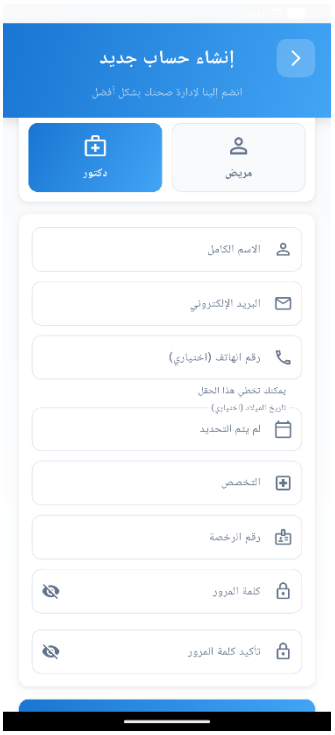


Figure 22 Doctor Registration Screen

Purpose:

The registration screen allows new users to create an account. Based on the selected role (Patient or Doctor), the form dynamically displays the appropriate fields. The process ensures all necessary information is collected while maintaining simplicity.

Layout & Interaction:

- ***Role Selection Tabs*** – at the top, two large buttons "Patient" and "Doctor". Tapping switches between the two registration forms. The active role is highlighted (usually with a different color or background).
    - **Image 3** shows the **Patient** form.
    - **Image 4** shows the **Doctor** form with additional professional fields.

- ***Common Fields (both roles):***
    - **Full Name** – required, text input with validation (minimum 3 characters, letters and spaces only).
    - **Email Address** – required, must be a valid email format; a real-time check for existing accounts is performed.
    - **Phone Number** – optional, with country code selector (default +20 for Egypt).
    - **Date of Birth** – optional, opens a date picker; user must be at least 13 years old.
    - **Password** – required, with a strength indicator (weak/medium/strong) based on length and character variety.
    - **Confirm Password** – required, real-time matching validation.

Doctor-Specific Fields:

- **Specialty** – required dropdown or text field (e.g., Cardiology, Dermatology).
- **License Number** – required, professional license identifier. *(These fields appear only when "Doctor" is selected, as seen in Image 4.)*

- **Create Account Button** – initially disabled until all required fields are valid. When pressed, it shows a loading spinner and submits the data. On successful registration, the user is navigated to the Email Verification screen (Section 4.5).
- **Login Link** – at the bottom, a text link "Already have an account? Login" that returns to the Login screen.

Flow:

1. User selects role (Patient or Doctor).
2. Fills in the required fields.
3. Clicks "Create Account".
4. If validation passes, the system creates the account, sends a verification email, and redirects to the Email Verification screen.
5. If any error occurs (e.g., email already in use), an appropriate error message is displayed.

4.5 Email Verification Screen



Figure 24 Verification Screen

Purpose:

Informs the user that a verification email has been sent and guides them through verification.

Components & Actions:

- Animated envelope icon (floating, pulsing).
- Instruction steps: (1) Open email app, (2) Find our message, (3) Click the verification link.
- **"Open Email App" button** – opens the device's email chooser.
- **"Resend Email" button** – disabled for 60 seconds after each send.
- **"Check Verification" button** – attempts to re-authenticate and verify the email status.
- If verified, a success animation plays and the user is redirected to the Login screen.

4.6 Patient Home Screen

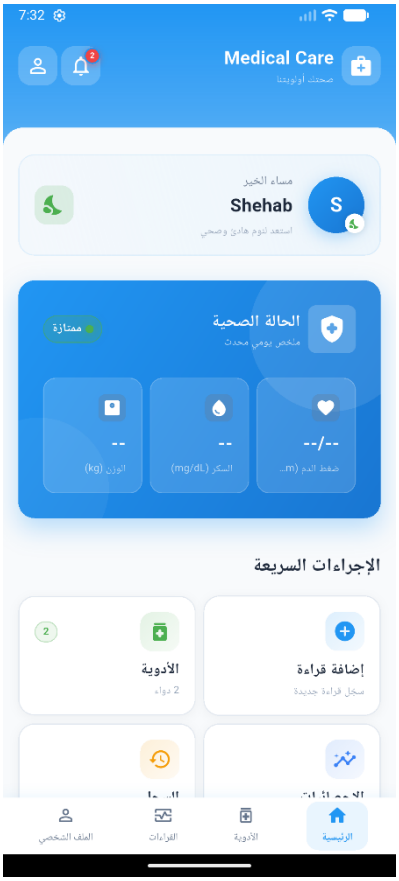


Figure 25 Patient Home Screen - 1

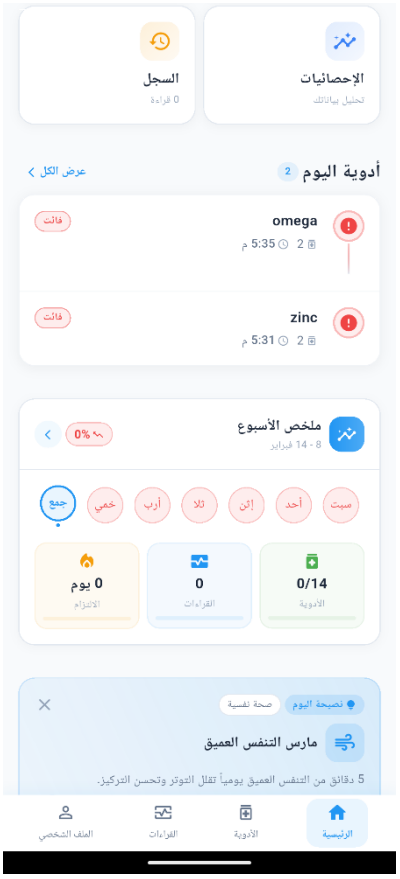


Figure 26 Patient Home Screen - 2

Purpose:

The central hub providing a snapshot of health status, quick actions, and today's medication schedule.

Sections & Functions:

- **Greeting Header** – time-based greeting with user name; tap on avatar to open Profile. Notification bell shows unread count and navigates to Notifications Center.
- **Health Status Card** – displays latest readings (BP, sugar, weight) with color-coded status; tap to go to Health Readings screen.
- **Quick Actions Grid** – four cards: "Add Reading", "Medicines", "Statistics", "History". Each tap navigates to the corresponding screen.
- **Today's Medicines** – lists upcoming/due medications with status badges. Each item has action buttons (Mark Taken, Skip, Snooze) when applicable. Buttons update the UI optimistically.

- **Weekly Progress Card** – shows adherence for the last seven days (colored day indicators) and statistics (medicines taken, readings recorded, streak days). Tap to view detailed statistics.
- **Daily Health Tip** – rotating tip card with share and dismiss options.

4.7 Medicine Management Screens

The Smart Medication Management System for Elderly People provides a comprehensive medicine management system, allowing patients to view all their active medications, track adherence, and easily add new medicines. This section consists of two main screens: the Medicine List screen and the Add/Edit Medicine screen.

4.7.1 Medicine List Screen

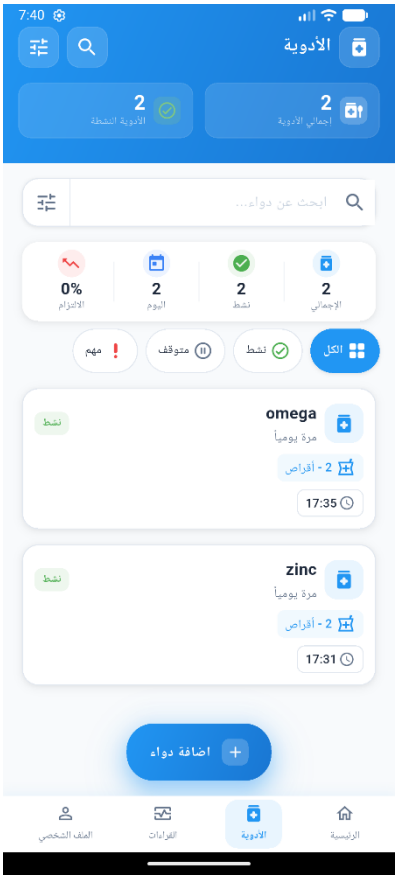


Figure 27 Medicine List Screen

Purpose:

Displays all medicines registered by the user, with search and filter capabilities, along with adherence statistics.

Components & Functions:

- **Top App Bar:**
  - Title "Medicines".
  - Search icon (🔍) to expand a search field.
  - Settings icon (⚙️) for future medicine settings.
- **Quick Statistics Card:**
  - **Active medicines count** (e.g., 2).
  - **Total medicines count** (e.g., 2).
  - **Adherence percentage** (e.g., 0%) with a small circular progress indicator.
- **Search Field:**
  - Appears when the search icon is tapped.
  - Real-time filtering of medicine names.
- **Medicine List:**
  Each item shows:
  - **Medicine name** (e.g., omega, zinc).
  - **Dosage & frequency** (e.g., once daily – 2 tablets).
  - **Scheduled time** (e.g., 17:35).
  - **Status** (active, paused, critical).
  - A status icon (e.g., exclamation mark for critical medicines).
  - Tapping an item opens the medicine details screen (edit/delete).
- **Floating Action Button:**
  - Large circular button at the bottom labeled "Add Medicine".
  - Tapping navigates to the Add Medicine screen.
- **Bottom Navigation Bar:**
  - Icons for main sections: Profile, Readings, Medicines, Home

4.7.2 Add New Medicine Screen

Purpose:

Allows the user to enter data for a new medicine or edit an existing one. The data is organized into several sections for clarity.

Section 1: Basic Information

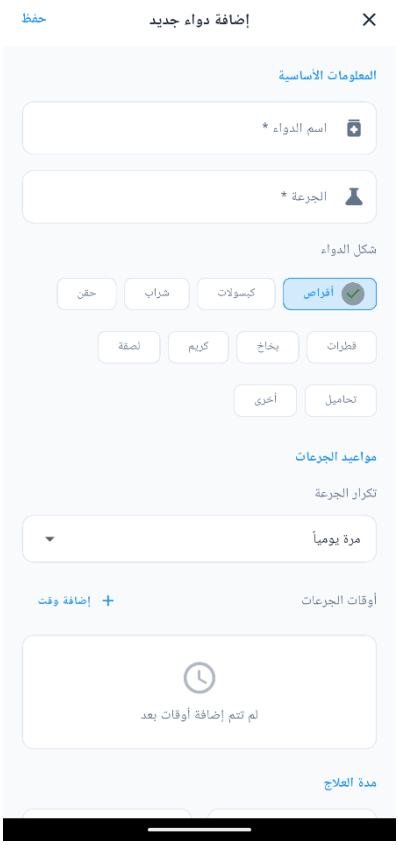


Figure 28 Basic Information

- **Medicine Name** – required field, with optional autocomplete suggestions.
- **Dosage** – required field, can be numeric or text (e.g., “2 tablets”).
- **Medicine Form** – selection buttons: Capsules, Injection, Syrup, Drops, Inhaler, Cream, Patch, Red (likely meaning tablet), Suppositories. The user selects the appropriate form.

Section 2: Dose Schedule

- **Dose Frequency** – dropdown (once daily, twice, three times, every day, etc.).
- **Dose Times** – multiple times can be added via the “Add Time” button. Added times are displayed in a list (e.g., 08:00, 20:00). If none added, a placeholder “No times added yet” appears.

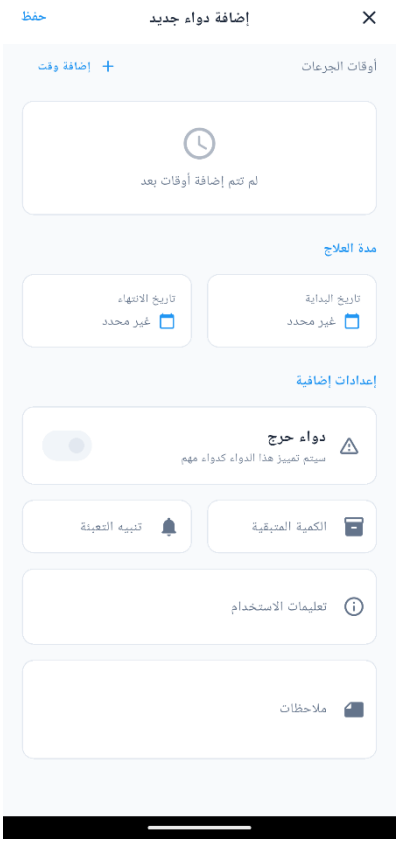


Figure 29 Dose Schedule

Section 3: Treatment Duration

- **Start Date** – date picker (default "Not set").
- **End Date** – date picker (can be left open for ongoing medication).

Section 4: Additional Settings

- **Critical Medicine** – switch; when enabled, the medicine is marked as important (shows a warning icon in the list).
- **Refill Reminder** – switch to enable notifications when stock is low.
- **Remaining Quantity** – optional numeric field for stock tracking.
- **Instructions** – optional multi-line text field.
- **Notes** – optional extra text field.

Save Button:

At the bottom, a "Save Medicine" button validates the inputs and saves the data to the database. After successful save, the user returns to the medicine list with the new medicine visible.

4.8 Health Readings Screens

The health readings module allows patients to record, view, and analyze their vital signs. It supports four types of measurements: **Blood Pressure**, **Blood Sugar**, **Weight**, and **Temperature**. The interface adapts to each type, providing appropriate input fields, validation ranges, and visual feedback.

4.8.1 Health Readings List Screen



Figure 30 Health Readings List Screen

Purpose:

Displays an overview of the latest readings with statistics and a chart for the selected measurement type and time period.

Components & Functions:

- **Top Bar:** Title “Health Readings” and a search icon.
- **Period Selector:** Tabs for Week, Month, Year, All. Switching the period updates the statistics and chart.
- **Type Tabs:** Four tabs: Blood Pressure, Blood Sugar, Weight, Temperature. The active tab is highlighted.
- **Statistics Card:** Shows the number of readings (e.g., “2 readings”) and key values:
    - For Blood Pressure: average, highest, lowest (e.g., 130/85 avg, 140/90 max, 120/80 min).

- For other types: average, min, max of the numerical value.

- **Chart Area:** An interactive line chart displaying the trend of the selected readings. The Y-axis scale adapts to the measured values (e.g., 120–200 mmHg for BP). Tapping on a data point reveals the exact value and time.
- **Legend:** Explains the colors used (e.g., systolic vs diastolic for BP, normal range shading).
- **Floating Action Button (FAB):** A large button at the bottom right with the label "Add New Reading" and the icon of the current type (e.g., a heart for BP). Tapping it navigates to the Add Reading screen with that type preselected.

4.8.2 Health Readings History Screen



Figure 31 Health Readings History Screen

Purpose:

Lists all recorded readings in chronological order, with status indicators and quick access to details.

Components & Functions:

- **Header:** Same as the list screen (title, search, period selector).
- **Reading List:**
  - ***Each item displays:***
    - **Value** with unit (e.g., “140/90 mmHg”).
    - **Status** (e.g., “High”, “Borderline”) with a colored background and icon.
    - **Relative time** (e.g., “just now”, “2 hours ago”).
  - Items are grouped by date (today, yesterday, etc.).

4.8.3 Select Reading Type Screen

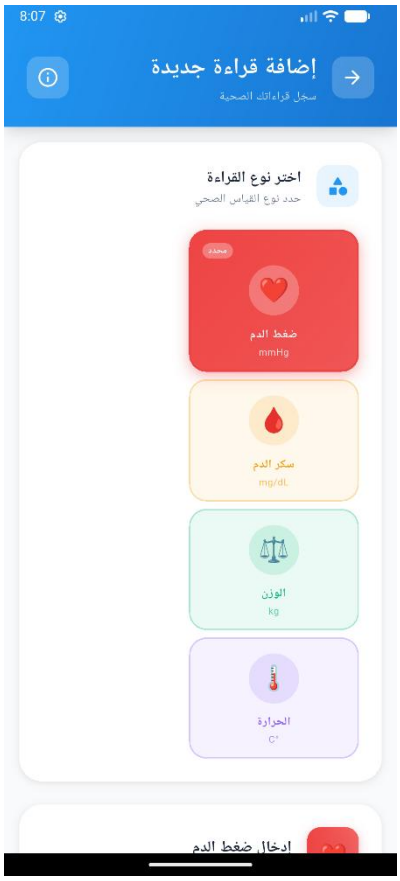


Figure 32 Reading Type

Purpose:

Appears when the user taps the FAB from the list screen (or when no type is preselected). Allows the user to choose which type of reading to add.

Components:

- **Header:** Title “Add New Reading” and subtitle “Choose the type of measurement”.
- **Type Grid:** Four large cards, each containing:
  - An emoji/icon (e.g., ❤️ for BP, 🩸 for sugar, ⚖️ for weight, 🌡️ for temperature).
  - The type name in Arabic and English.
  - The unit (mmHg, mg/dL, kg, °C).

4.8.4 Add Health Reading Screen



Figure 33 Add Health Reading

Purpose:

Allows the user to enter a new measurement. The layout and input fields change according to the selected type. **The same screen structure is used for all types, with only the input fields and validation rules adapting.**

General Layout (for all types):

- **Header:** Shows the selected type's name and a brief instruction.
- **Input Section:**
  - For **Blood Pressure:** Two separate numeric fields for systolic (upper) and diastolic (lower), each with a placeholder and unit. A hint shows the acceptable range (e.g., "40–300 mmHg").
  - For **Blood Sugar, Weight, Temperature:** A single numeric field with unit. A range hint is also displayed.
- **Measurement Time (for Blood Sugar only):** A dropdown or chips to indicate the context (fasting, after meal, etc.). This is optional.
- **Notes Field:** Optional multi-line text for additional comments.
- **Quick Actions (Preset Values):** A row of chips with common values for the selected type (e.g., "120/80", "130/85", "140/90" for BP). Tapping a chip automatically fills the input field(s) with that value.
- **Preview Card:** As the user types, a real-time preview shows the interpreted value, its health status (normal, warning, critical) with a colored badge, and any advice (e.g., "Your blood pressure is normal"). For weight, if the user's height is known, the BMI is calculated and displayed here.
- **Save Button:** A large, colored button at the bottom. When pressed, it validates the input, checks rate limits, and saves the reading. A success dialog appears afterward, offering to add another reading or return to the list.

Important Note: The same screen is used for all four reading types. The only differences are:

- The **input fields** (single vs. double).
- The **unit** displayed.
- The **quick action presets.**
- The **validation ranges** and **status calculation** (which are handled by the backend logic, not the UI).

This unified approach reduces code duplication and provides a consistent user experience across all health metrics.

4.9 Profile Screens

The profile section provides users with a centralized place to view their personal information, track account statistics, manage settings, and access support. The screens are designed to be clean and organized, with clear categories and intuitive navigation.

4.9.1 Profile Overview Screen

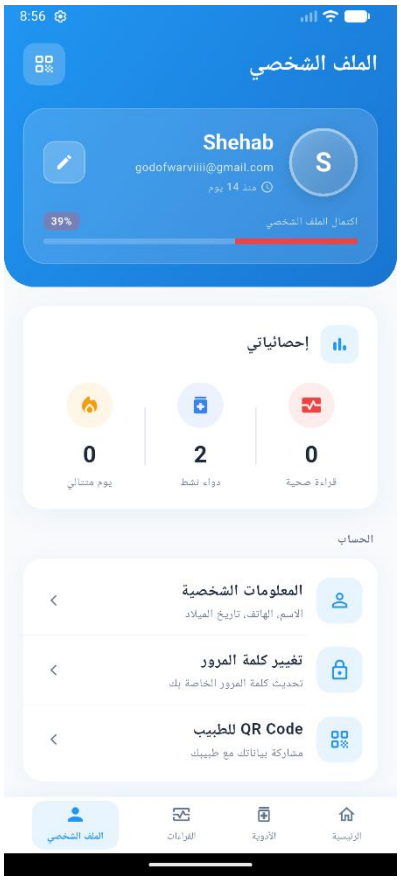


Figure 34 Profile Overview

Purpose:

Displays the user's basic information, profile completion progress, key statistics, and links to account management features.

Components & Functions:

- **Profile Header:**
  - User avatar (initials or photo).
  - User name (e.g., "Shehab").
  - Email address (e.g., "godofwarvili@gmail.com").
  - Account age (e.g., "Since 14 days").

- **Profile Completion Progress:**
  - A circular progress bar showing the percentage of profile completeness (e.g., 39%).
  - Encourages the user to fill in missing information.
- **Statistics Cards:**
  - **Streak days** (e.g., 0) – number of consecutive days with activity.
  - **Active medicines** (e.g., 2) – count of currently active medications.
  - **Health readings** (e.g., 0) – total number of recorded readings.
  - Tapping any card may navigate to the relevant section.
- **Account Section:**
  - **Personal Information** – leads to a screen where the user can edit name, phone, date of birth, etc.
  - **Change Password** – opens the password change flow.
  - **Doctor QR Code** – displays a QR code containing the user's medical ID or profile link, which can be scanned by a doctor to quickly access the patient's data (planned feature).

4.9.2 Personalization & Support Screen

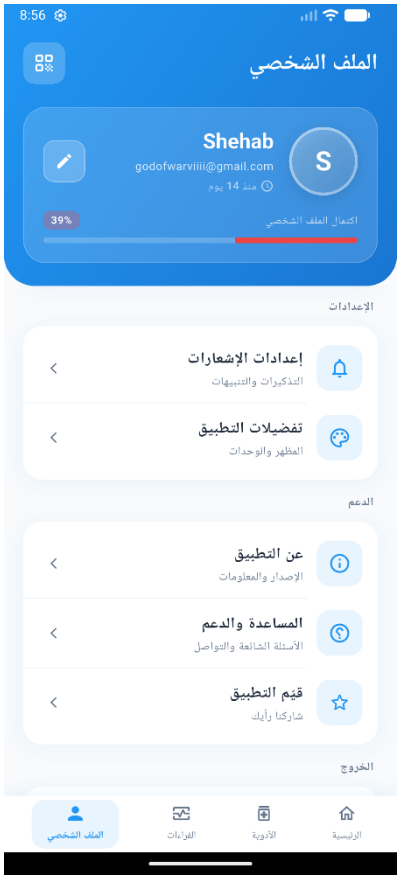


Figure 35 Personalization \ Support Screen

Purpose:

Groups together settings related to app behavior and user support.

Components & Functions:

- **Profile Header** (same as previous screen).
- **Personalization Settings:**
  - **Notification Settings** – configure which notifications to receive (medicine reminders, appointment alerts, health tips).
  - **App Preferences** – choose theme (light/dark), measurement units, language, etc.
- **Support Section:**
  - **About App** – displays version number, developer information, and legal notices.
  - **Help & Support** – opens a FAQ or contact form.
  - **Rate the App** – opens the store page for rating.

- **Logout Button:**
  A prominent red button at the bottom that signs the user out.

4.9.3 Account Actions Screen (Image 14)



Figure 36 Account Actions Screen

Purpose:

Provides advanced account management options, including logout and account deletion.

Components & Functions:

- **Profile Header** (same as previous).
- **Support Section** (same as previous screen, possibly redundant but included).
- **Logout Section:**
  - **Sign Out** – logs the user out and returns to the login screen.
  - **Delete Account** – initiates a destructive flow with confirmation dialogs to permanently remove the user's data.

4.10 Doctor Screens

The doctor module allows healthcare providers to manage linked patients, scan patient QR codes, view patient lists, and monitor medical reports. The doctor screens follow the same application design style, using a blue medical theme, clear cards, simple icons, and a bottom navigation bar.

4.10.1 Doctor Dashboard Screen



Figure 37 Doctor Dashboard Screen

The Doctor Dashboard Screen is the main screen for doctors after login. It displays the title **"لوحة الطبيب"** and provides quick access to doctor functions.

The screen includes a doctor information card, a large **"مسح كود المريض"** card for scanning the patient QR code, and a section for **"أحدث المرضى"**. This helps the doctor quickly link patients and access recent patient records.

Main features:

- Doctor profile summary.
- Patient QR code scanning.

4.10.2 Patients List Screen

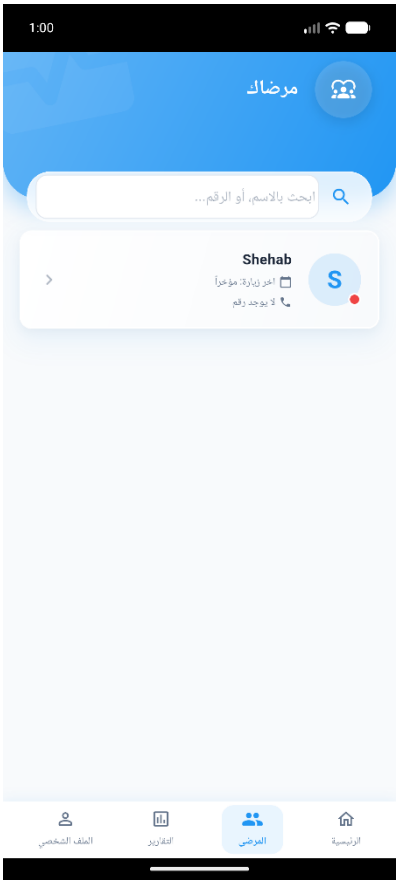


Figure 38 Patients List Screen

The Patients List Screen displays all patients linked to the doctor. It contains a search bar that allows the doctor to search by patient name or number.

Each patient card shows basic information such as patient name, last visit, phone status, and a navigation arrow to open patient details.

Main features:

- Search patients by name or number.
- View linked patients.
- Display last visit information.
- Open patient details.
- Bottom navigation bar.

4.10.3 Medical Reports and Analytics Screen



Figure 39 Doctor Medical Reports / Analytics Screen

The Medical Reports Screen provides summarized analytics for the doctor. It includes filters for **7 days**, **30 days**, and **90 days**.

The screen shows important statistics such as total patients, active medicines, critical alerts, and average medication adherence. It also includes a medication adherence ranking section and a button to export reports as PDF.

Main features:

- Medical analytics overview.
- Time period filter.
- Total patients.
- Active medicines.
- Critical alerts.
- Average adherence.
- Medication adherence ranking.
- Export PDF report.

Chapter 5: Implementation

This chapter presents the implementation details of the Smart Medication Management System for Elderly People, focusing on the adopted software architecture, user interface design principles, and the achieved system outcomes. The implementation phase translates the proposed design concepts into a fully functional, secure, and scalable mobile application.

5.1 Software Architecture



Figure 40 Software Architecture

The Smart Medication Management System for Elderly People is built upon a robust, enterprise-grade software architecture that strictly adheres to the principle of **Separation of Concerns (SoC)**. A **Feature-First (Modular) Architecture**, integrated with the core principles of **Clean Architecture**, was adopted to ensure high scalability, maintainability, and testability.

This architectural approach enables independent feature development, reduces coupling between system components, and enhances long-term system evolution.

5.1.1 Structural Organization

The codebase is systematically organized into clearly defined layers and modules:

- **Core Module**
  This module contains the foundational infrastructure of the application, including global configuration files, centralized error handling mechanisms, theme definitions (design tokens), and hardware-level services such as device security checks and connectivity monitoring.
- **Feature Modules**
  Each major application feature (e.g., Authentication, Doctor Dashboard, Health Readings, and Medicines) is encapsulated within its own module. Every feature module includes its presentation layer (screens), state providers, and feature-specific business logic, effectively preventing unwanted coupling between unrelated functionalities.
- **Data and Services Layers**
  Centralized directories manage domain models (such as UserModel) and external service integrations.

5.1.2 State Management and Routing

- **State Management**
  The application employs the **Provider** package to implement reactive state management. Both global providers (e.g., AuthProvider and SettingsProvider) and local providers (e.g., MedicineFormProvider) are used strategically to optimize performance. This approach ensures that only the affected portions of the widget tree are rebuilt when state changes occur.
- **Routing System**
  A type-safe and declarative navigation system was implemented using **GoRouter**. The routing architecture integrates authentication guards that prevent unauthorized access to protected routes. Additionally, it supports custom animated page transitions (such as slide and fade effects), enhancing the overall user experience and navigation flow.

5.1.3 Backend Infrastructure and Security

The backend infrastructure is fully powered by **Firebase** services. A key architectural achievement is the implementation of advanced **Firestore Security Rules**, enforcing strict **Role-Based Access Control (RBAC)** between Patients and Doctors.

To mitigate malicious activities, rate-limiting mechanisms were introduced using dedicated collections (e.g., login_attempts), alongside immutable audit logs (security_logs) for security traceability. Furthermore, the application supports resilient offline behavior through a robust **100 MB offline persistence cache**, ensuring data continuity during network disruptions.

5.2 User Interface Design

The User Interface (UI) was designed with a strong emphasis on **Human–Computer Interaction (HCI)** principles, aiming to provide an intuitive, accessible, and reassuring experience for both patients and healthcare professionals.

5.2.1 Design System and Visual Consistency

A strict design system was implemented using centralized UI constants, including AppTheme, AppColors, and AppTextStyles. This token-based design methodology guarantees visual consistency across the application. Typography hierarchy, color schemes, shadows, and spacing all originate from a single source of truth, making the UI scalable and easily adaptable for future re-theming.

5.2.2 Role-Tailored User Experiences

The application interface dynamically adapts based on the authenticated user role:

- **Patient Dashboard**
  Focuses on vital health tracking, scheduled medication reminders, and personalized health alerts.

- **Doctor Dashboard**
  Doctor Dashboard provides tools for patient management, secure access to medical records through QR-code linking, medication adherence monitoring, and medical analytics reports.

5.2.3 Error Handling and User Feedback

Continuous system status feedback is prioritized within the UI. For example, an OfflineBanner is non-intrusively displayed when network connectivity is lost, keeping users informed without blocking interactions. Additionally, customized UI states such as ErrorScreen and ComingSoonScreen gracefully handle broken navigation paths and unavailable features, preventing dead-end user experiences.

5.3 Results and Discussion

The implementation phase successfully validated the Smart Medication Management System design, resulting in a stable, secure, and feature-rich mobile application.

5.3.1 Technical Achievements

One of the most significant achievements is the advanced **Background Notification System**. By integrating flutter_local_notifications with background execution handlers (NotificationActionHandler), users can interact directly with medication reminders from the device notification tray—even when the application is fully terminated (cold state). Pending actions are reliably detected and synchronized with Firestore upon the next application launch.

5.3.2 System Resilience and Reliability

Robust error handling was implemented at the root execution level using runZonedGuarded and PlatformDispatcher. When combined with **Firebase Crashlytics**, all unhandled runtime exceptions are captured, ensuring application stability and minimizing crash occurrences.

Furthermore, device integrity checks were performed to detect rooted or jailbroken devices. Upon detection, sensitive caching functionalities were disabled, reinforcing the system's commitment to high security standards.

5.3.3 Overall Impact

In conclusion, the Smart Medication Management System for Elderly People operates exactly as intended. The seamless integration of native device capabilities—such as biometrics, time zone handling, and QR code scanning—with a cloud-based, serverless backend resulted in a highly efficient architecture. The system demonstrates a strong capability for securely managing real-time patient–doctor interactions and digitizing personal health records. Moreover, the architecture is scalable and well-prepared to support future hospital-level extensions.



Figure 41 System Resilience Workflow

Chapter 6: Testing

This chapter presents the testing strategy adopted to validate the Smart Medication Management System for Elderly People. Software testing is a critical phase in the software development lifecycle, as it verifies that the implemented system behaves as expected and helps identify defects before deployment. In the context of a healthcare application, testing is particularly essential because inaccurate data handling or unreliable system behavior may lead to incorrect medical information being presented to patients and doctors.

To ensure correctness, reliability, and security, the Smart Medication Management System for Elderly People was evaluated through two main testing levels: **unit testing** and **integration testing**. Unit testing was used to validate individual components in isolation, while integration testing was conducted to verify the interaction between system modules, including the user interface, state management, routing, backend services, and security rules.

6.1 Unit Testing

Unit testing focuses on verifying the smallest testable components of the application in isolation. Each test evaluates a specific function, method, model, or helper class to ensure that it produces the expected output for a given input. In the Smart Medication Management System for Elderly People, unit tests were applied to four major domains: **data models**, **authentication validators**, **health reading validators**, and **input sanitization utilities**.

6.1.1 Data Model Tests

Data models form the backbone of the application, as all information exchanged between the user interface, business logic, and Firestore database passes through these structures. Therefore, validating model construction, serialization, deserialization, helper methods, and edge-case handling was necessary to ensure data consistency and reliability.

Three core models were tested in this domain:

- UserModel
- Medicine

- HealthReading

6.1.1.1 UserModel Tests

The UserModel represents both patients and doctors and includes more than 25 fields. Unit tests were designed to verify object creation, role-specific field handling, serialization and deserialization behavior, helper properties such as age calculation and profile completeness, and equality behavior. The results confirmed that the model handles both required and optional fields correctly while preserving data integrity across application layers.

6.1.1.2 Medicine Model Tests

The Medicine model manages medication records, including validation, sanitization, scheduling-related attributes, and lifecycle operations. Tests were conducted to verify object creation, field sanitization, serialization compatibility, logical state transitions such as deletion and expiration, and reminder-related helper methods. The results demonstrate that the model safely handles both valid and edge-case medication data.

6.1.1.3 HealthReading Model Tests

The HealthReading model is responsible for storing and classifying vital sign measurements, including blood pressure, blood sugar, and temperature. The unit tests verified reading creation, medical status classification, value clamping, display formatting, advice generation, and comparison methods. The results confirm that the model correctly identifies normal, abnormal, and critical readings based on the implemented medical thresholds.

6.1.2 Authentication Validator Tests

The AuthValidators class acts as the first defensive layer for user registration and login. It validates user input before any data reaches the authentication service or the database. The validation rules cover general identity information and medical-specific fields, including:

- Email addresses
- Passwords and password strength
- Names and phone numbers
- Blood types, height, weight, and date of birth

- Doctor-specific fields such as specialty, license number, and years of experience

6.1.3 Health Reading Validator Tests

The HealthReadingValidator class was tested to ensure that medically invalid readings are rejected before storage. It uses a type-safe validation design based on a sealed class structure (ValidationResult, ValidationSuccess, and ValidationFailure). The test cases evaluated missing values, out-of-range values, logical inconsistencies, optional notes handling, and input safety against malicious content. The results confirm that only medically and structurally valid readings are accepted by the system.

6.1.4 Input Sanitizer Tests

The InputSanitizer class is used to clean user inputs before storage and processing. Since the application accepts free-text notes and some numeric fields from users, sanitization was required to reduce the risk of injection attacks and malformed content. Unit tests verified the removal of HTML and



Figure 42 Input Sanitizer Testing Process

JavaScript fragments, whitespace normalization, truncation of overlong notes, safe parsing of numeric values, and handling of null or empty inputs. The results indicate that the sanitization layer performs reliably and contributes to data security.

6.1.5 Unit Testing Summary

A total of **109 unit test cases** were executed across the tested domains. The results are summarized in the table below.

Test Domain	Test Cases	Passed	Failed	Pass Rate
Data Models (UserModel, Medicine, HealthReading)	45	45	0	100%
Authentication Validators	40	40	0	100%
Health Reading Validators	15	15	0	100%
Input Sanitizer	9	9	0	100%
Total	**109**	**109**	**0**	**100%**

Table 2 Unit Testing

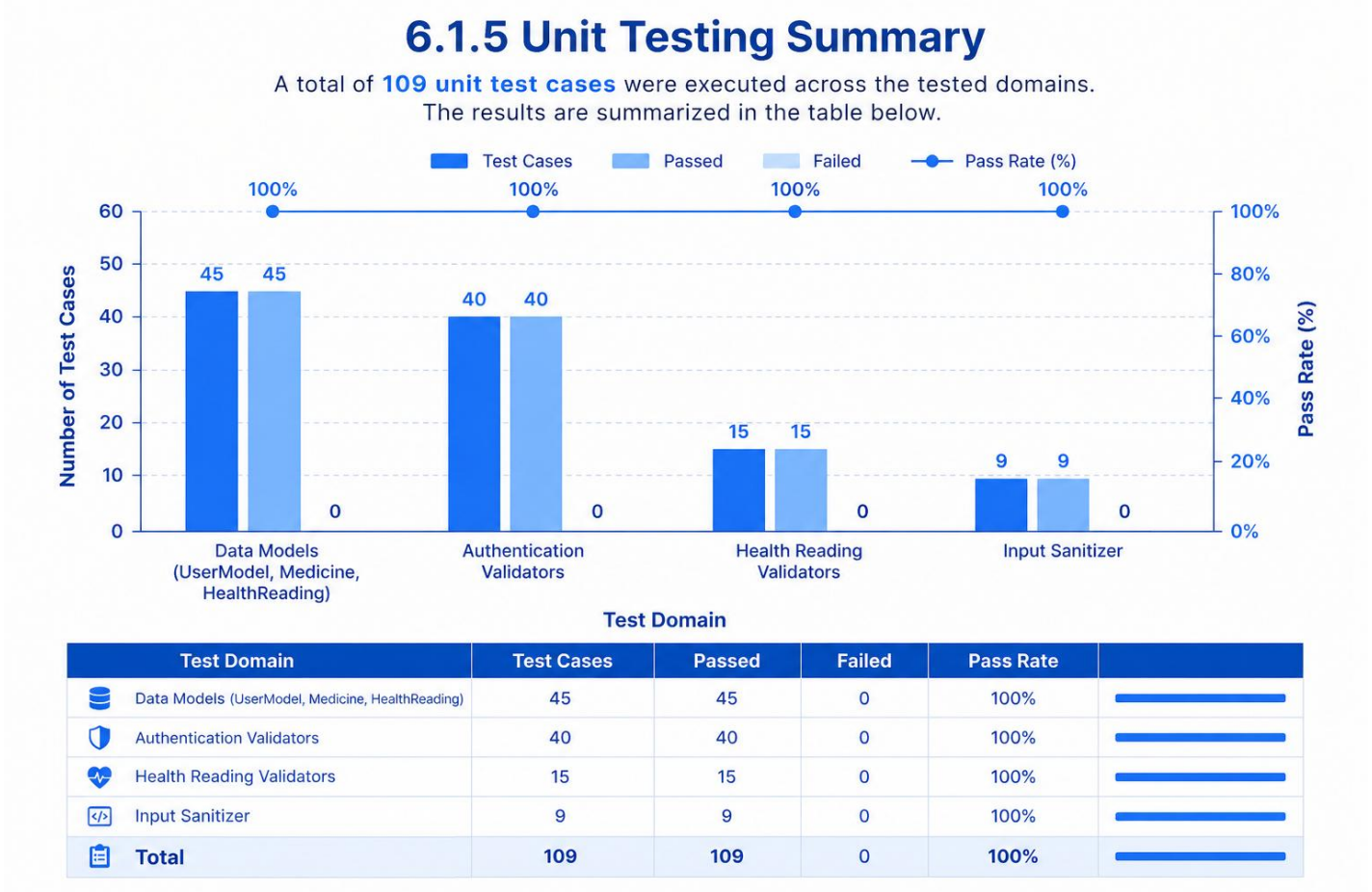


Figure 43 Unit Testing Summary

All **109 unit tests** passed successfully. The comprehensive coverage of boundary values, null handling, type coercion, sanitization, and medically sensitive validation rules provides strong evidence that the core business logic of the Smart Medication Management System for Elderly People is robust, secure, and functionally correct.

6.2 Integration Testing

Integration testing was performed to verify that multiple components operate correctly as a unified system. Unlike unit tests, which isolate individual classes or methods, integration tests validate full workflows involving the user interface, state management (Provider), navigation (GoRouter), backend services (Firebase), local notifications, offline behavior, and Firestore security rules.

The integration testing strategy focused on the following domains:

- Authentication workflows
- Navigation and routing behavior
- Medicine management
- Notification lifecycle handling
- Offline resilience and recovery
- Firestore security rule enforcement

6.2.1 Authentication Flow Integration Tests

The authentication subsystem involves multiple coordinated components, including AuthProvider, AuthService, AuthValidators, Firebase Authentication, and Firestore. Integration tests were conducted to validate the complete authentication lifecycle, including patient and doctor registration, verified and unverified login scenarios, logout behavior, password reset functionality, email verification cooldown mechanisms, authentication state persistence, and retry handling under temporary network failure. The results confirm that the authentication workflow behaves correctly and enforces the intended access control and user-state transitions.

6.2.2 Navigation and Routing Integration Tests

The routing system, implemented using GoRouter, was tested to ensure that authentication guards and role-based navigation behave as intended. The tests verified redirection rules for authenticated and unauthenticated users, deep-link navigation through notifications, role-dependent dashboard

rendering, and graceful handling of invalid routes. These results confirm that protected navigation paths are enforced reliably and that the application remains user-friendly even in exceptional routing scenarios.

6.2.3 Medicine Management Integration Tests

The medicine management workflow combines user forms, state providers, Firestore CRUD operations, and notification scheduling. Integration tests were performed to verify complete end-to-end scenarios, including adding medication records, deleting medicines using soft-delete logic, assigning medicines to patients through doctor workflows, and detecting expired medication plans. The results confirm that medication data is stored, updated, and synchronized correctly across the system.

6.2.4 Notification System Integration Tests

The notification subsystem represents one of the most complex areas of the application. It integrates LocalNotificationService, MedicineReminderService, NotificationActionHandler, and NotificationScheduler. Integration tests validated the scheduling of reminders, user actions such as **Mark Taken**, **Skip**, and **Snooze**, processing of pending actions after cold start, and Firebase Cloud Messaging behavior in both foreground and background states. These results confirm that reminder notifications are delivered reliably and that user actions are correctly reflected in the application state and backend records.

6.2.5 Offline and Resilience Integration Tests

Given that healthcare applications may be used in unstable network conditions, offline resilience was treated as a critical quality attribute. Integration tests were conducted to verify Firestore offline caching behavior, visibility of the OfflineBanner, lifecycle-based refresh behavior on resume, and restricted caching when a compromised device is detected. The results demonstrate that the application remains usable during network loss while preserving its security policies under potentially unsafe execution environments.

6.2.6 Integration Testing Summary

A total of **42 integration test cases** were executed across the six major workflow domains. The results are summarized below.

Test Domain	Test Cases	Passed	Failed	Pass Rate
Authentication Flow	10	10	0	100%
Navigation and Routing	7	7	0	100%
Medicine Management	4	4	0	100%
Notification System	7	7	0	100%
Offline and Resilience	4	4	0	100%
Firestore Security Rules	10	10	0	100%
Total	**42**	**42**	**0**	**100%**

Table 3 Integration Testing

All **42 integration tests** passed successfully. These results confirm that the Smart Medication Management System for Elderly People functions correctly as an integrated system and that its security, routing, notification, and offline workflows are implemented reliably.

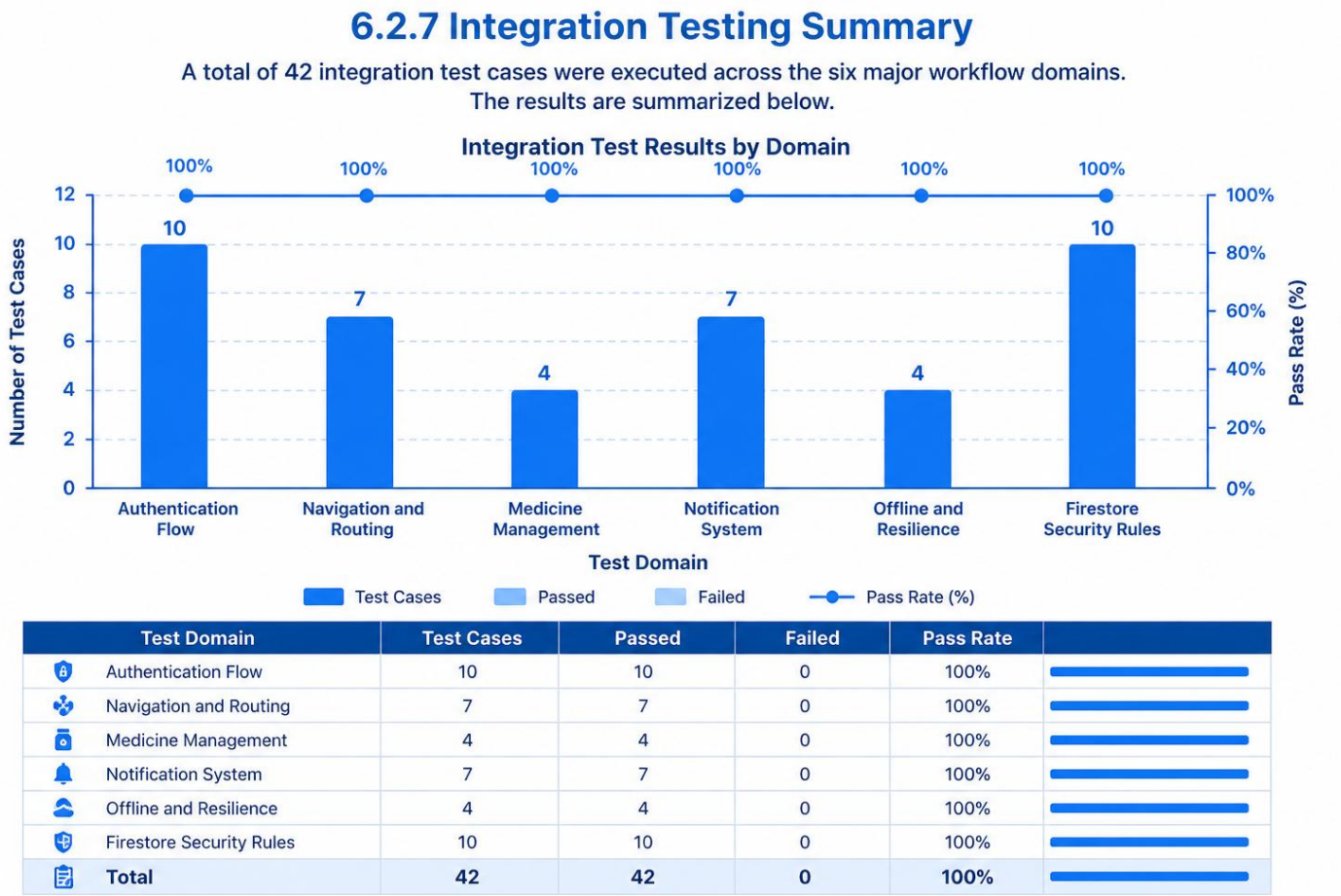


Figure 44 Integration Testing Summary

6.3 Overall Testing Summary

The testing phase of the Smart Medication Management System for Elderly People included both unit testing and integration testing to ensure correctness at the component level and reliability at the system level. In total, **151 test cases** were executed.

Testing Level	Total Tests	Passed	Failed	Pass Rate
Unit Testing	109	109	0	100%
Integration Testing	42	42	0	100%
Grand Total	**151**	**151**	**0**	**100%**

Table 4 Overall Testing

The comprehensive testing strategy—covering data models, validation rules, sanitization utilities, authentication workflows, routing guards, notification lifecycles, offline resilience, and Firestore security rules—provides strong confidence that the Smart Medication Management System for Elderly People satisfies its functional requirements, security constraints, and reliability objectives. The **100% pass rate** across all test domains demonstrates a high level of implementation maturity and indicates that the application is well-prepared for deployment and future extension.

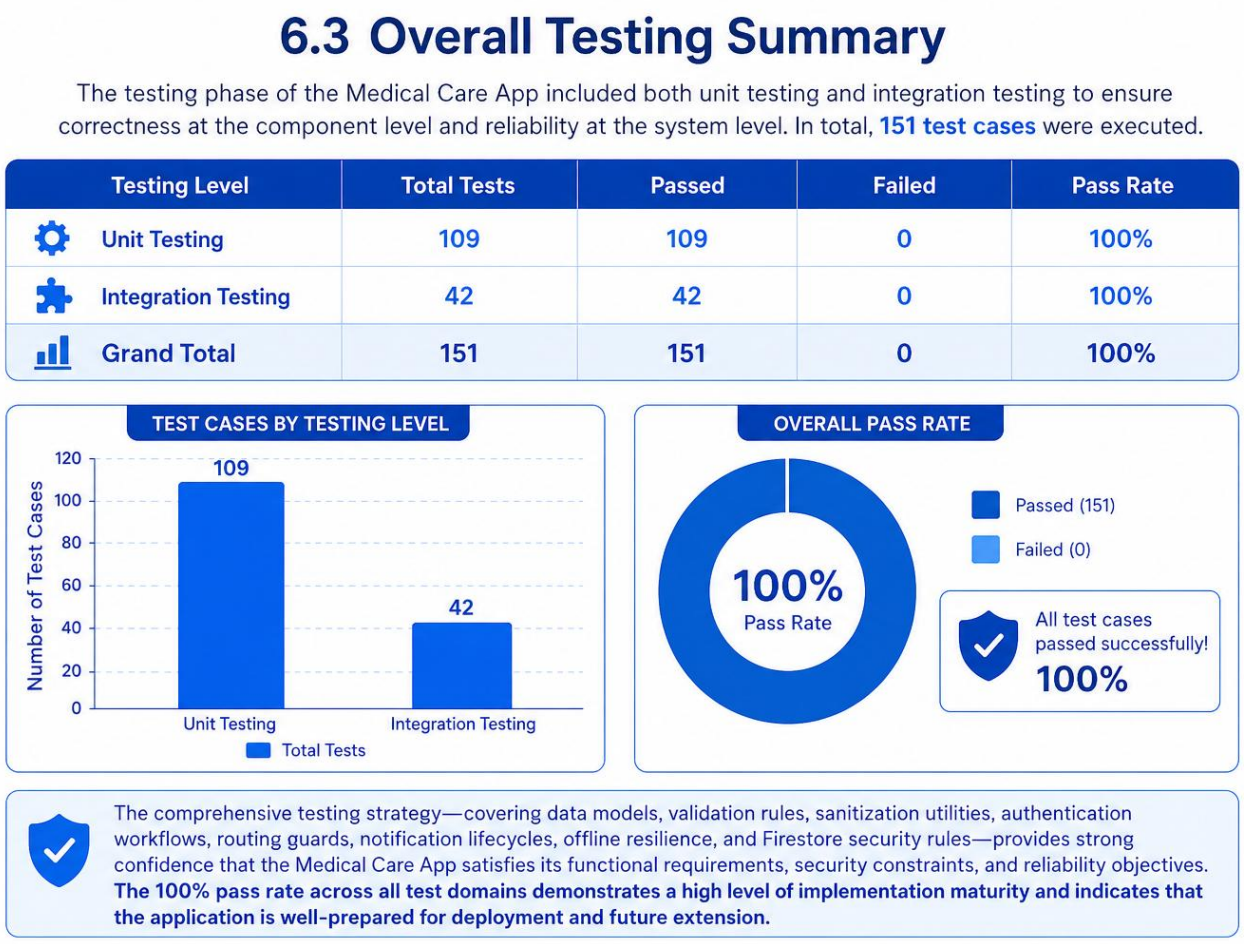


Figure 45 Overall Testing Summary

Conclusion

The Smart Medication Management System for Elderly People represents a comprehensive effort to address a real-world challenge in the healthcare domain: the fragmentation and limited accessibility of personal health records. Throughout this project, a cloud-based mobile application was designed, developed, and tested to enable patients to manage their health information—including vital sign readings, medication schedules, and dose tracking—while also allowing doctors to monitor and prescribe treatments within a secure digital ecosystem.

The system was implemented using **Flutter** as a cross-platform development framework and supported by a **serverless Firebase backend**, including Authentication, Cloud Firestore, Cloud Storage, Cloud Messaging, Analytics, and Crashlytics. This technology stack enabled rapid development while maintaining a high level of scalability, reliability, and maintainability.

One of the key strengths of the project lies in its **Feature-First Modular Architecture**, which was combined with the principles of **Clean Architecture**. This architectural approach produced a well-structured, extensible, and maintainable codebase in which each major feature—such as Authentication, Medicines, Health Readings, Doctor Dashboard, Notifications, Profile, and Settings—operates as an independent module.

Another major achievement is the implementation of a strong **Role-Based Access Control (RBAC)** mechanism at both the application layer and the database layer through carefully designed **Firestore Security Rules**. This security model ensures that patient records remain accessible only to authorized users and protects the system against several security risks, including unauthorized access, field tampering, injection attempts, and repeated failed login attacks through rate-limiting and immutable audit logs.

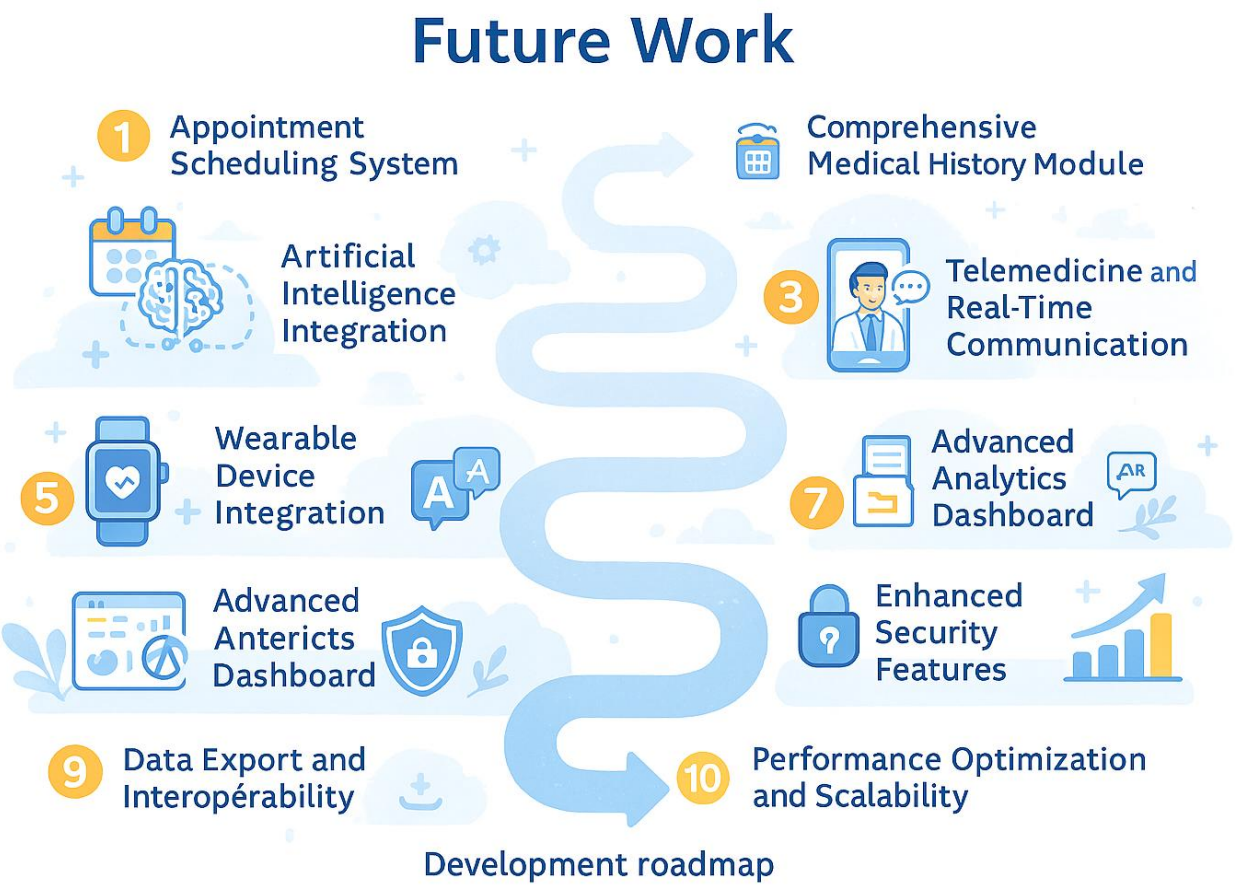
In addition, the **medicine reminder notification system** represents one of the most technically advanced components of the project. The system supports scheduled local notifications with interactive actions such as **Mark Taken**, **Skip**, and **Snooze**. It also handles notification responses even when the application is fully terminated and synchronizes pending actions with the cloud database once the application is reopened. This ensures continuity in medication adherence tracking across different application states.

The application also demonstrated strong **resilience and reliability** under adverse conditions. Through Firestore's offline persistence capabilities,

users can continue accessing critical health information even during network interruptions. Furthermore, device integrity verification was incorporated to detect rooted or jailbroken devices, and sensitive caching operations were restricted when device compromise was detected, thereby strengthening overall security.

From a quality assurance perspective, the project was validated through a comprehensive testing strategy that included both **unit testing** and **integration testing**. A total of **151 test cases** were executed successfully with a **100% pass rate**, providing strong evidence of the system's correctness, reliability, and security readiness.

In conclusion, the Smart Medication Management System for Elderly People successfully achieved its primary objectives. It provides a secure and user-friendly platform for digitizing personal health records, supporting patient–doctor interaction through QR-based linking, automating medication adherence through intelligent reminders, and delivering meaningful health insights through real-time vital sign classification. Overall, the project demonstrates that modern cross-platform technologies can be effectively used to build a practical, scalable, and deployment-ready healthcare solution.



Although the Smart Medication Management System for Elderly People has successfully achieved its core objectives, several opportunities remain for future enhancement and expansion. The following directions are proposed for future development:

1. Appointment Scheduling System

A complete appointment management module can be implemented to allow patients to book, reschedule, and cancel appointments with their linked doctors. This feature may also include automated reminders, calendar synchronization, and doctor-side availability management.

2. Comprehensive Medical History Module

Future versions of the system may provide a chronological medical history view that includes diagnoses, laboratory results, surgical records, and prescription history. Such a feature would improve long-term patient monitoring and support clinical decision-making.

3. Artificial Intelligence Integration

Machine learning techniques can be introduced to analyze patient data and generate predictive insights. Examples include anomaly detection in vital

sign trends, medication adherence prediction, and personalized health recommendations based on historical readings and patient profiles.

4. Telemedicine and Real-Time Communication

The addition of real-time communication features—such as in-app messaging, voice calls, or video consultations—would significantly enhance the practical value of the system by enabling direct interaction between patients and doctors.

5. Wearable Device Integration

Future development may include integration with wearable devices and health platforms such as **Apple HealthKit**, **Google Health Connect**, and Bluetooth-enabled medical devices. This would allow automatic health data collection and reduce reliance on manual entry.

6. Expanded Multi-Language Support

Although the current system supports Arabic and English, future work could extend localization support to additional languages commonly used in the target region, thereby improving accessibility for a wider user base.

7. Advanced Analytics Dashboard

A more advanced dashboard for doctors can be developed to provide aggregated insights across patients, including treatment adherence rates, health trends, and interactive visual reports to support clinical assessment.

8. Data Export and Interoperability

Future releases may support standardized data export options such as **PDF**, **CSV**, and **HL7 FHIR** formats. This would improve interoperability with external healthcare systems and simplify record sharing across institutions.

9. Enhanced Security Features

Additional security improvements may include:

- **Multi-Factor Authentication (MFA)** for stronger account protection
- **End-to-End Encryption** for highly sensitive medical data
- **Session Management** to allow users to monitor and revoke active sessions

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ملخص فكرة المشروع

يهدف مشروعنا إلى تقديم تطبيق طبي سهل وآمن يساعد كبار السن والمرضى على تنظيم أدويتهم ومتابعة حالتهم الصحية بصورة منتظمة. جاءت فكرة المشروع نتيجة لما يواجهه كثير من المرضى من صعوبة في تذكّر مواعيد تناول الدواء، خصوصًا عند استخدام أكثر من علاج خلال اليوم، مما قد يؤدي إلى نسيان الجرعات أو تناولها في مواعيد غير صحيحة، بالإضافة إلى صعوبة الاحتفاظ بسجل منظم للقياسات الصحية ومشاركته مع الطبيب

يسمح التطبيق للمستخدم بإضافة بيانات الأدوية، مثل اسم الدواء، والجرعة، وعدد مرات الاستخدام، ومواعيد تناولها، ثم يقوم بإرسال إشعارات وتنبيهات في الأوقات المحددة لمساعدة المريض على الالتزام بالخطة العلاجية. كما يمكن للمستخدم تسجيل أهم القياسات الصحية، مثل ضغط الدم، ومستوى السكر، والوزن، ودرجة الحرارة، مع الاحتفاظ بسجل للقراءات السابقة، مما يساعد على متابعة تطور الحالة الصحية بمرور الوقت

يعتمد النظام على التخزين السحابي لمزامنة بيانات المستخدم وحفظها بصورة آمنة، مع إمكانية الاحتفاظ ببعض البيانات على الهاتف عند عدم توافر اتصال بالإنترنت، ثم مزامنتها عند عودة الاتصال. كما يدعم التطبيق مشاركة البيانات الطبية من خلال رمز QR Code مما يسمح للطبيب بالوصول إلى معلومات المريض وقراءة سجل الأدوية والقياسات الصحية بطريقة سريعة ومنظمة

تم تصميم واجهات التطبيق لتكون واضحة وسهلة الاستخدام ومناسبة لكبار السن، من خلال تقليل الخطوات المطلوبة للوصول إلى الوظائف الأساسية، واستخدام عناصر مرئية واضحة، بالإضافة إلى دعم اللغتين العربية والإنجليزية. ويطبق النظام مجموعة من إجراءات الأمان، مثل إنشاء حساب، وتسجيل الدخول، والتحقق من البريد الإلكتروني، وتحديد صلاحيات الوصول، وحماية البيانات الطبية من الاستخدام غير المصرح به

يساهم المشروع في تقليل أخطاء تناول الأدوية، وتحسين التزام المرضى بمواعيد العلاج، وتوفير سجل صحي منظم يساعد الأطباء على متابعة الحالات واتخاذ قرارات أفضل. كما يمكن تطوير النظام مستقبلًا بإضافة حجز مواعيد الأطباء، والتواصل عن بُعد، وربط التطبيق بالأجهزة القابلة للارتداء، واستخدام الذكاء الاصطناعي لتحليل القراءات الصحية والتنبيه المبكر عند اكتشاف أي تغيرات غير طبيعية

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